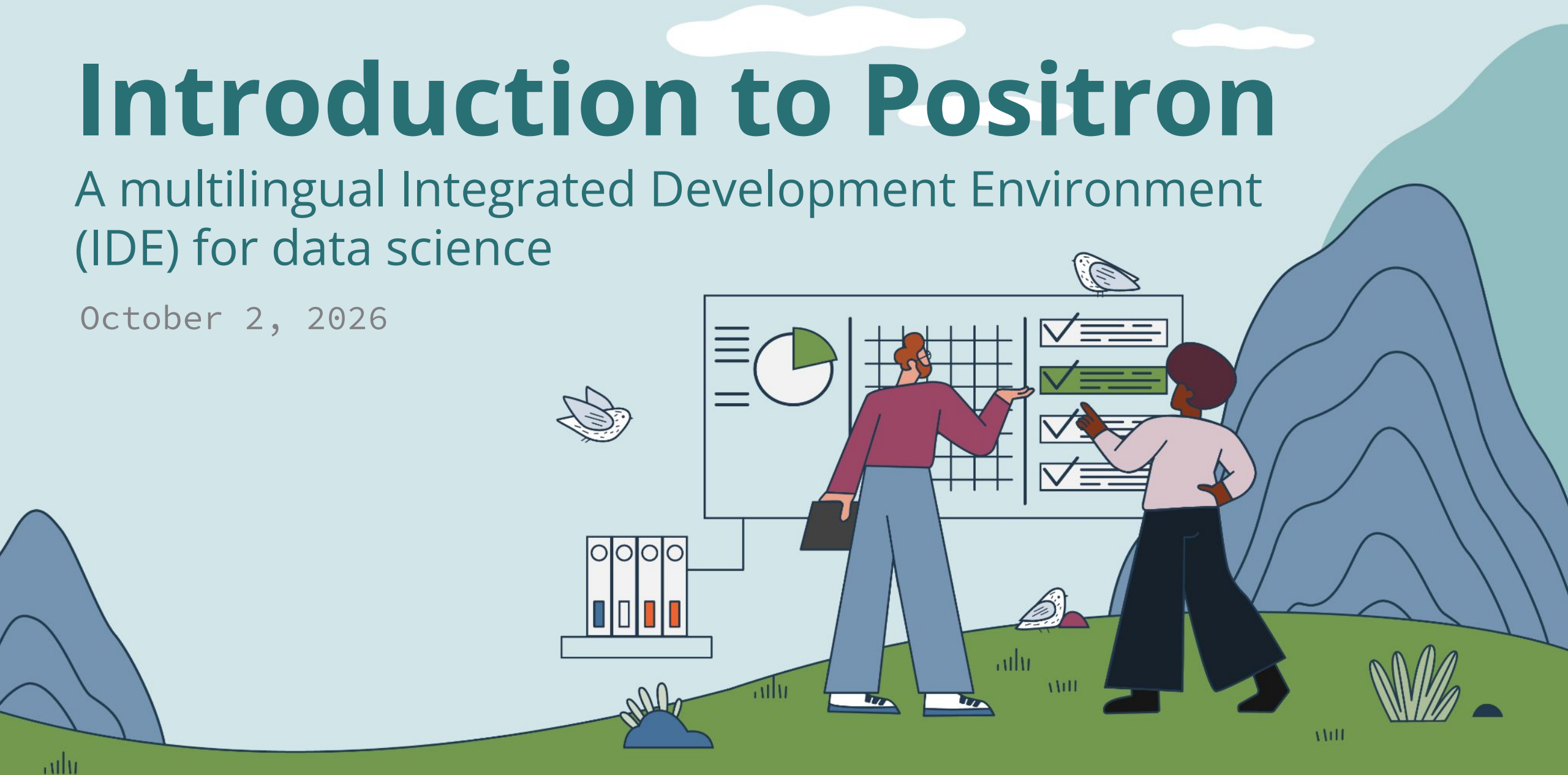


Introduction to Positron

A multilingual Integrated Development Environment (IDE) for data science

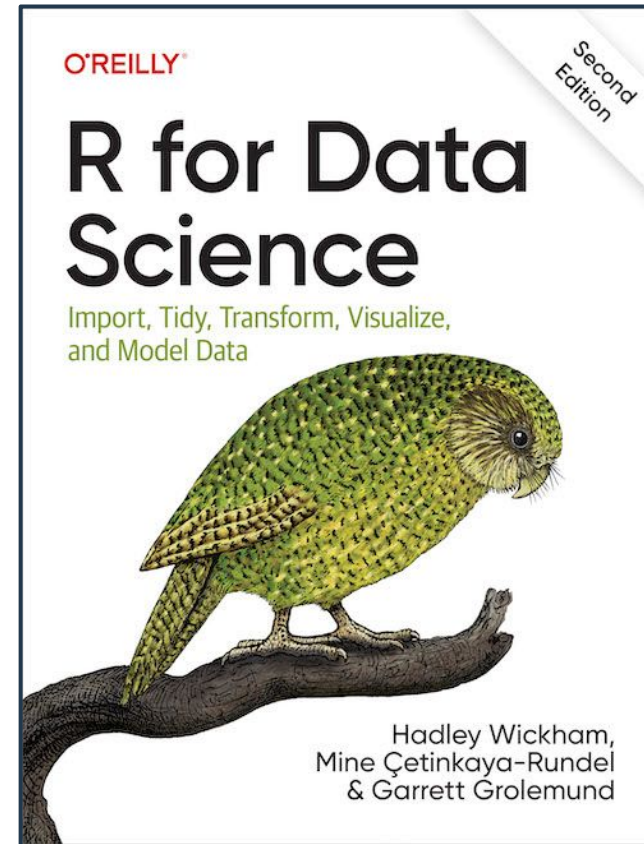
October 2, 2026



Your Instructor



Garrett Grolemond
Director of Learning
Posit PBC



Posit Academy

FREE

Self-paced, on-demand learning for every Posit user. No cost. Live now at academy.posit.co



Solutions

Products

Open Source

Customers

Resources

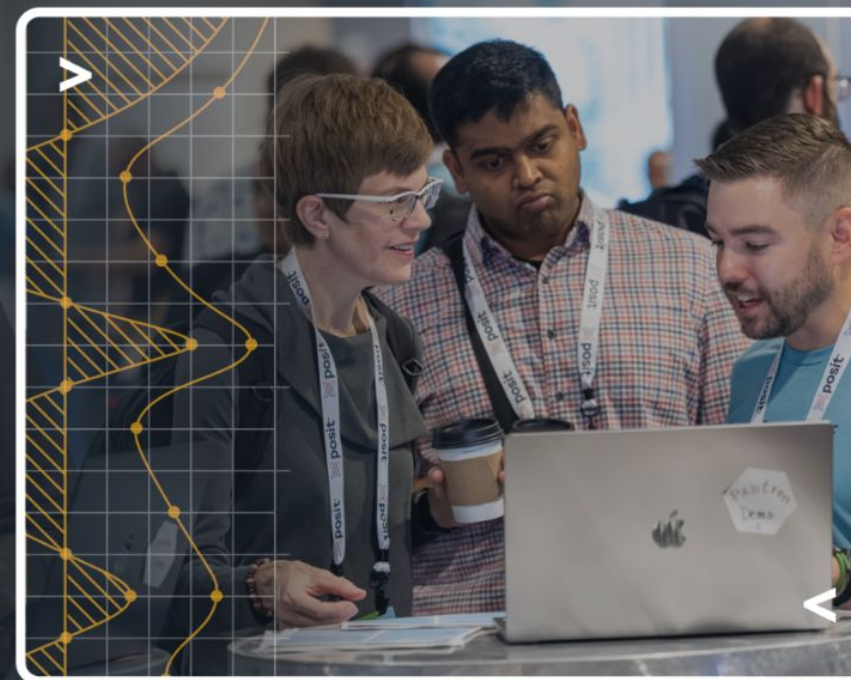
Search



Welcome to Posit Academy

Your learning hub for data science with R and Python — courses, live trainings, and a thriving community.

FULL COURSE CATALOG



Get Help

posit::conf(2026) is coming to Houston, TX! Join us Sept 14–16.

RSVP NOW

Learning Objective

Enable every participant to begin using Positron today for data science workflows in R.

1. **Introduction to Positron**

- RStudio transition to Positron
- Customizing Positron Settings
- R Projects and Packages
- File management
- Git Integration

2. **AI with Posit Assistant**

- Introduction to Posit Assistant
- Data exploration with Posit Assistant
- Coding with Posit Assistant
- Context management
- Commands
- Skills

3. **Clinical Data Wrangling and Analysis**

- Exploring AdaM data
- Building plots and tables with the DT package
- Publishing reports
- Introduction to Quarto

Introduction to Positron



Today's materials



Will remain available after the workshop at
<https://github.com/posit-dev/positron-pharma-quickstart>

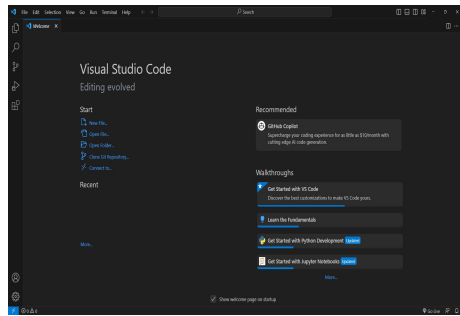
Log into Workbench



Positron VS Code adapted to data science

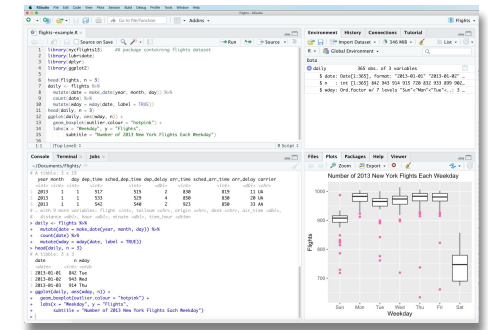
POSITRON

VS CODE

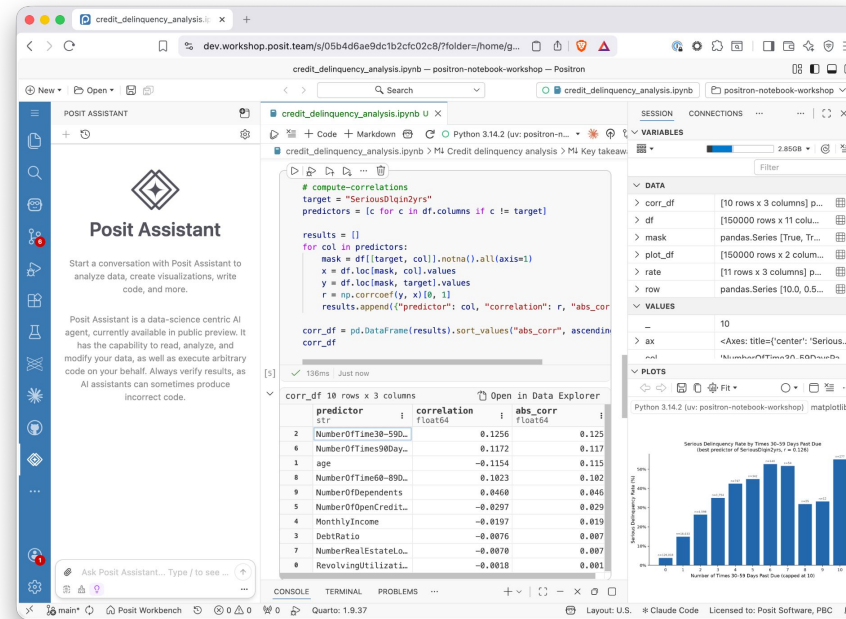


Software Developers

RSTUDIO



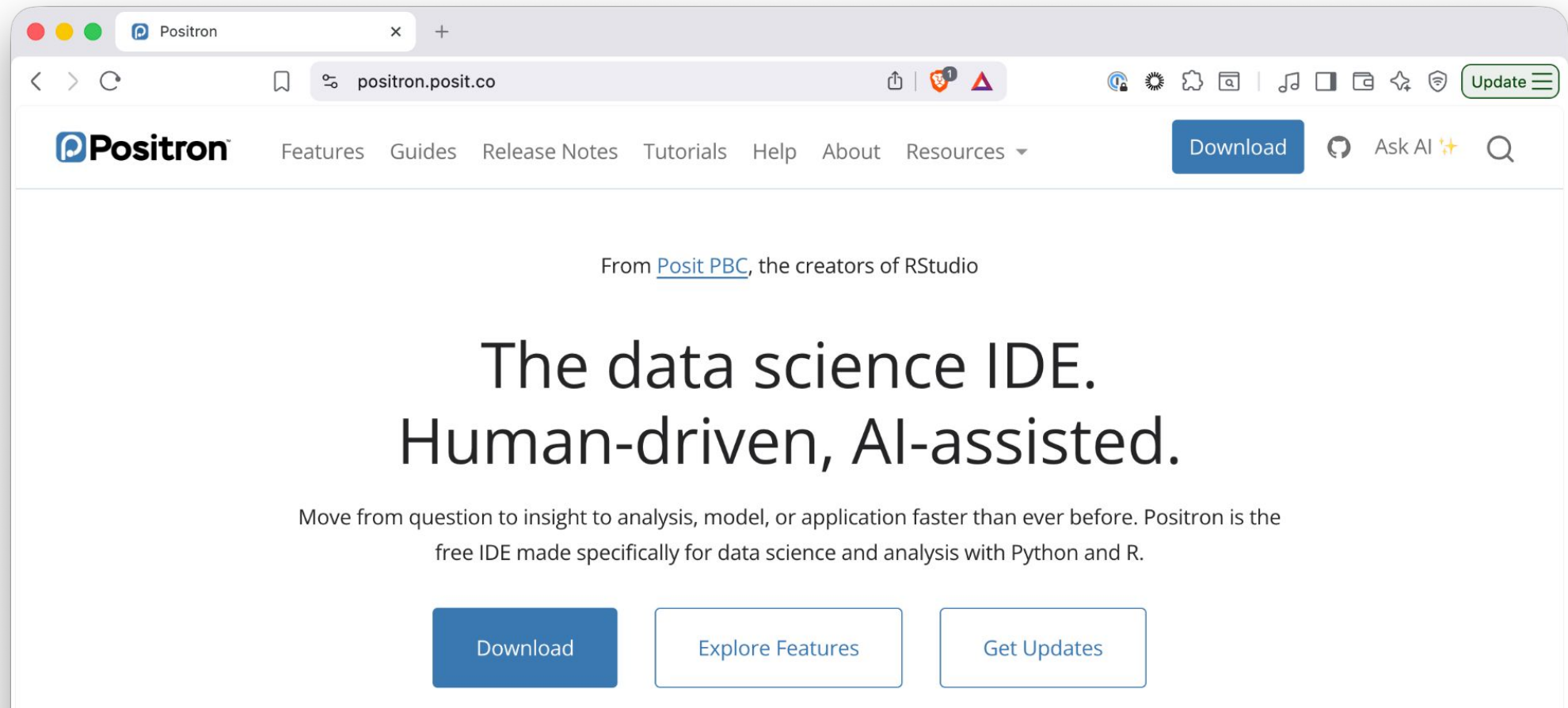
Data Scientists



Data Scientists

Documentation at

positron.posit.co



Security

Make the model secure:

- HIPAA-compliant BAA with the model provider
- A local model in a compliant environment

Or make the data secure:

- Remove Personal Identifier Information
- No AI in environments with sensitive data

This is a bad idea.

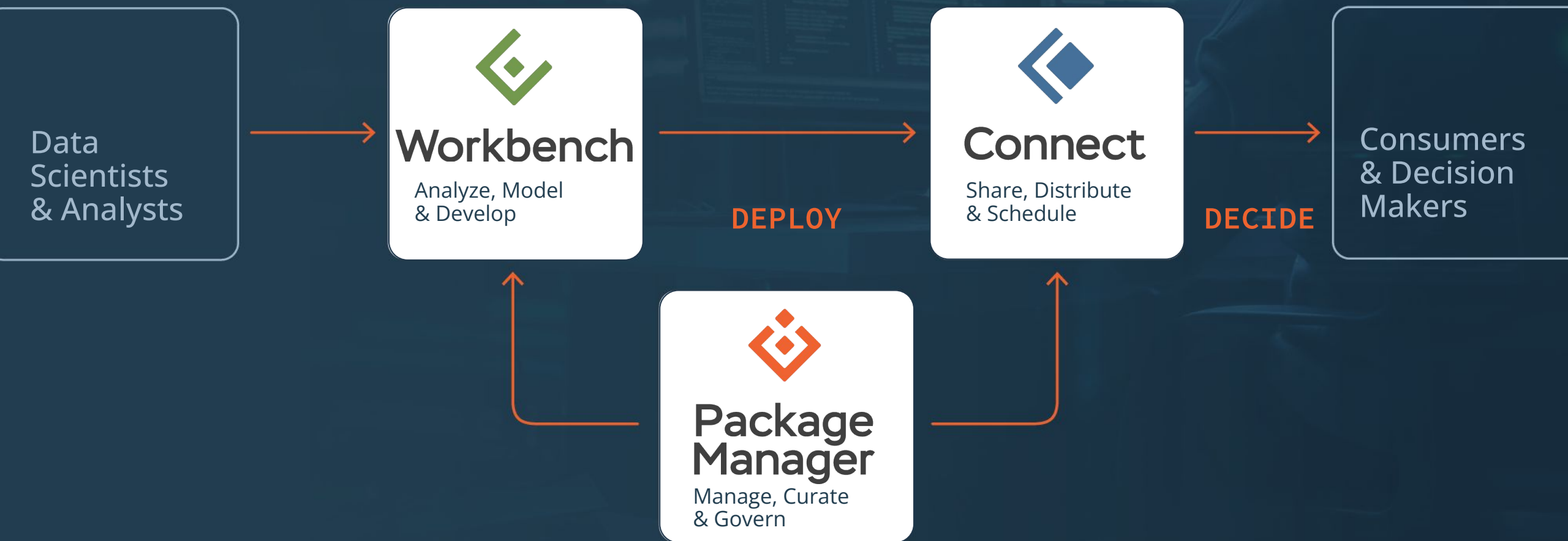
An unsafe
LLM

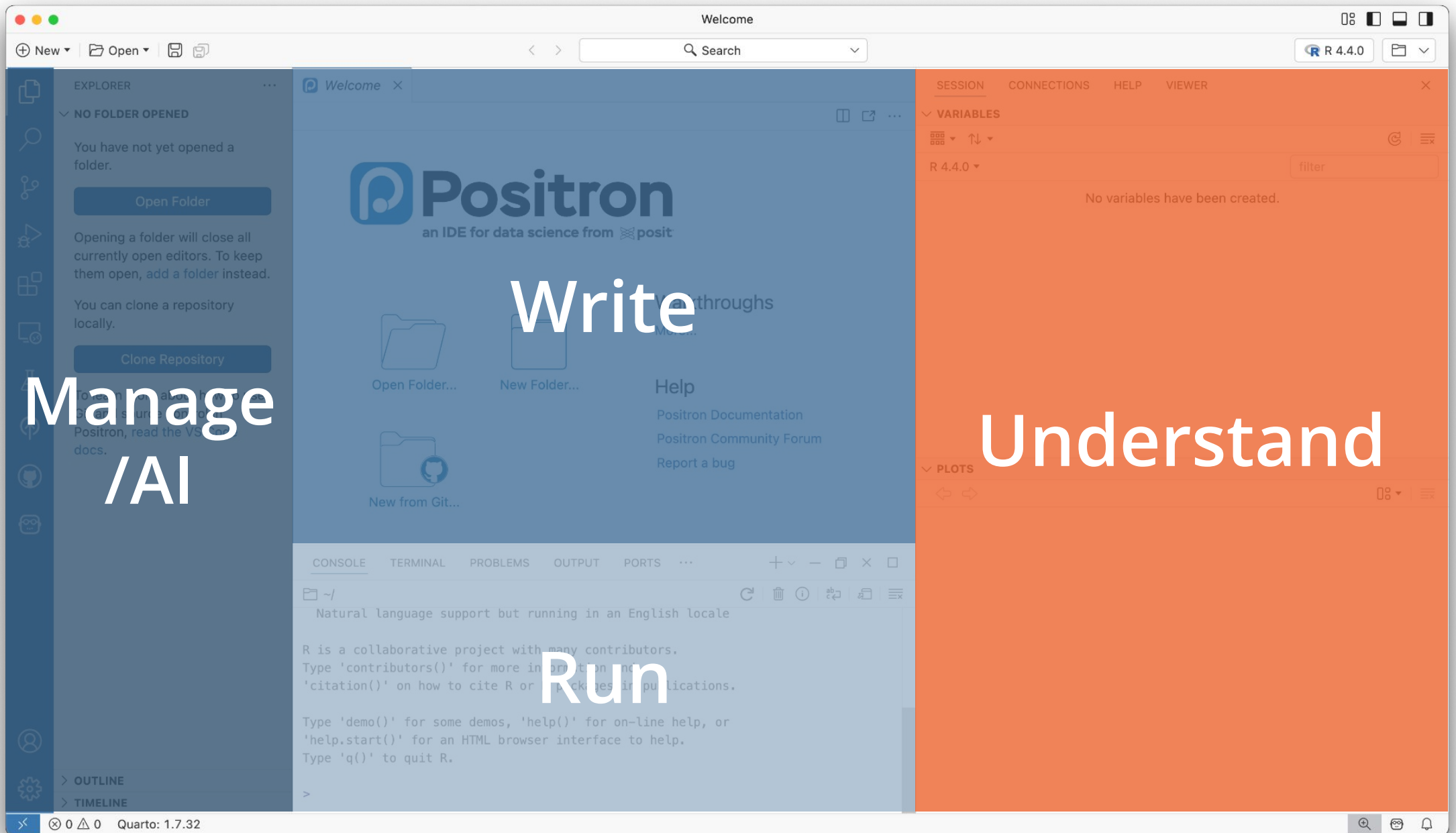
Your IT
safeguards



Posit Team - A Professional Insight Suite

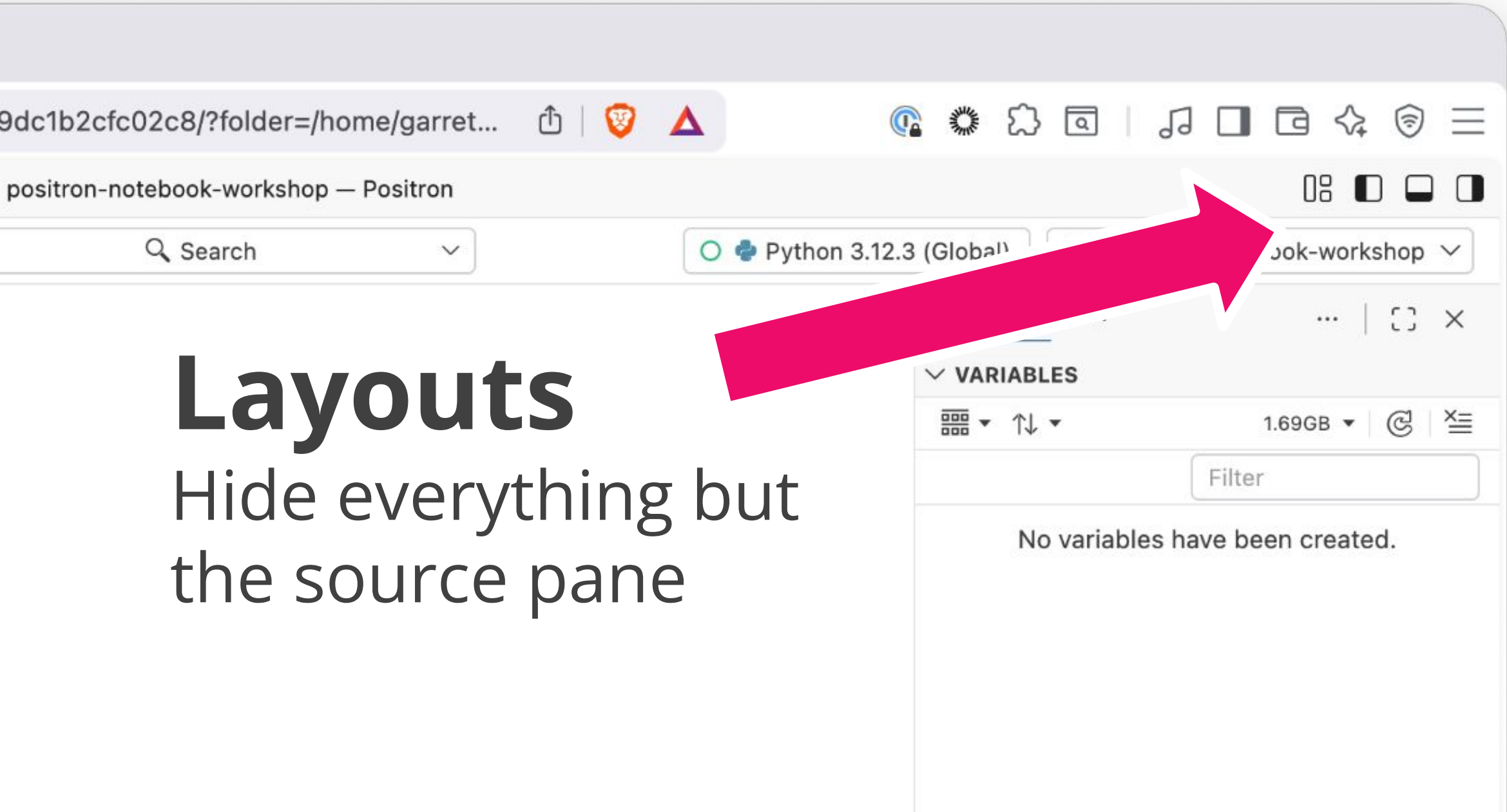
A complete intelligence stack for code-first data science and research. Data scientists can work safely, independently, and quickly, without IT handoffs or hand-holding.





Differences from RStudio IDE





Layouts

Hide everything but
the source pane

Projects = Folders

New Folder From Template

Python Environment

How would you like to set up your Python environment?

- ☒ Create a new virtual environment (Recommended)
- ☐ Use an existing environment

Environment Creation

Select a way to create a new virtual environment

uv Use uv to find a Python version locally (or install it if needed) for the new environment

Environment name

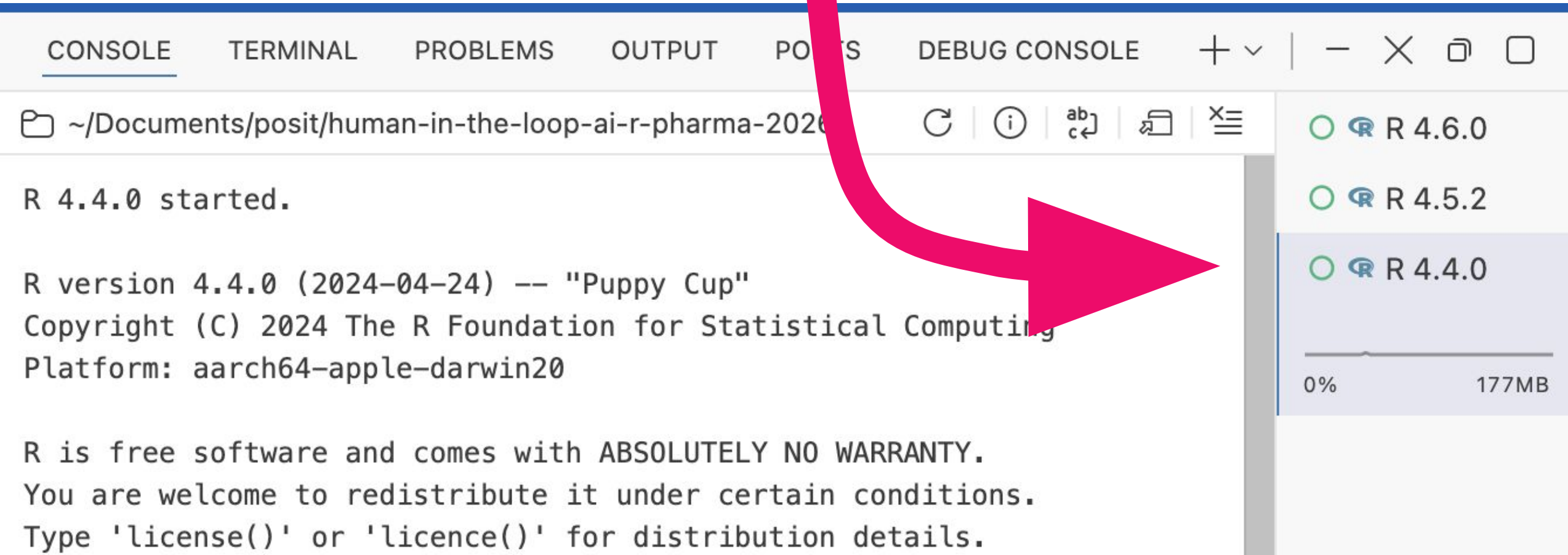
.venv

New >

New Folder from Template

Multiple Consoles

Let you switch between different versions of R



The screenshot shows the RStudio interface with the following components:

- Top Bar:** Contains tabs for `CONSOLE`, `TERMINAL`, `PROBLEMS`, `OUTPUT`, `POINTER`, and `DEBUG CONSOLE`. To the right are icons for adding, removing, and toggling console windows.
- File Explorer:** Shows the current project path: `~/Documents/posit/human-in-the-loop-ai-r-pharma-2020`.
- Console:** Displays the output of the R session, including the version and platform information.

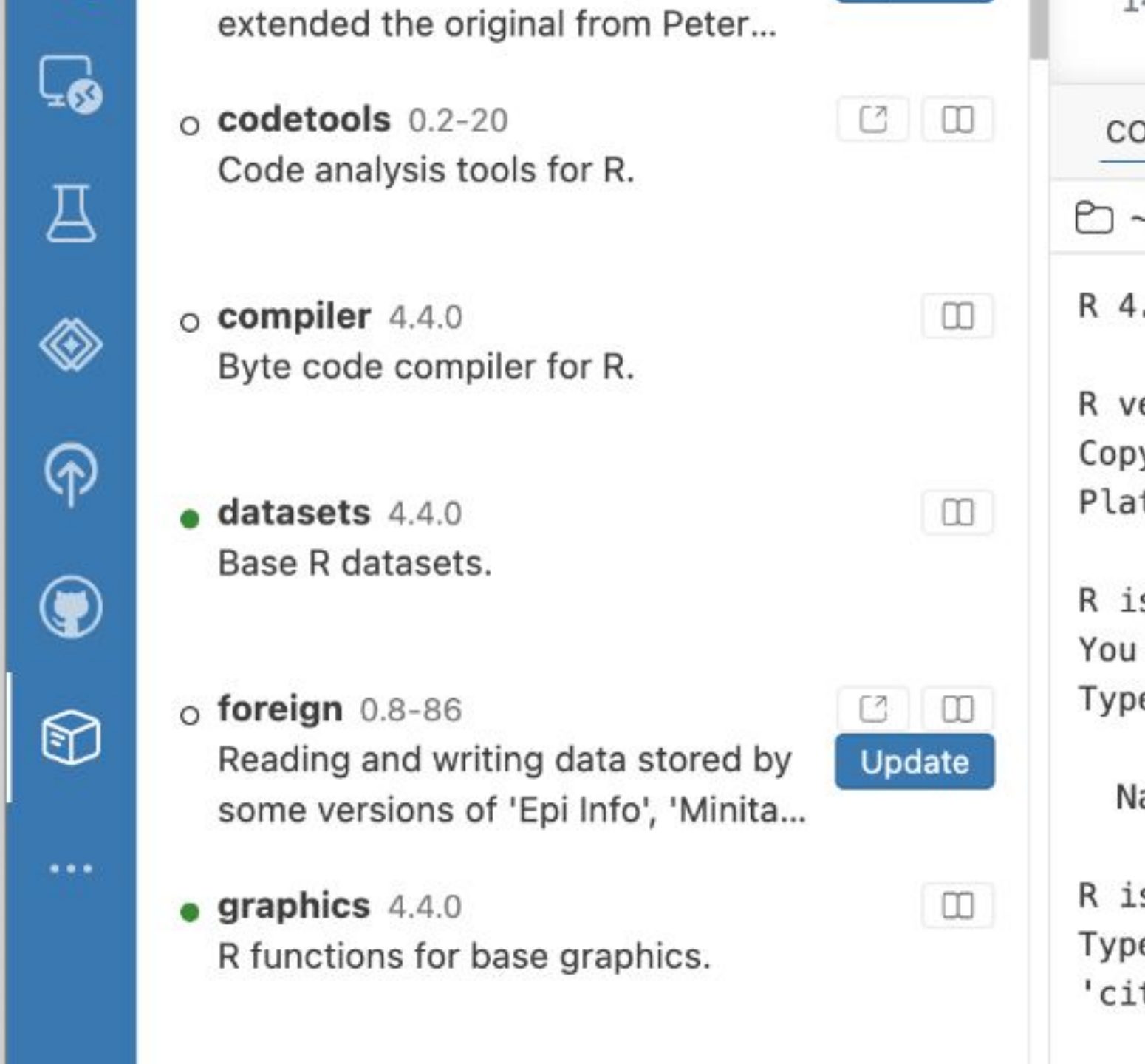
```
R 4.4.0 started.  
  
R version 4.4.0 (2024-04-24) -- "Puppy Cup"  
Copyright (C) 2024 The R Foundation for Statistical Computing  
Platform: aarch64-apple-darwin20  
  
R is free software and comes with ABSOLUTELY NO WARRANTY.  
You are welcome to redistribute it under certain conditions.  
Type 'license()' or 'licence()' for distribution details.
```
- R Version Selector:** Located on the right side, it lists the installed R versions: `R 4.6.0`, `R 4.5.2`, and `R 4.4.0`. The `R 4.4.0` version is currently selected and highlighted.

Positron does not save state

R --no-save --no-restore-data

Packages

Moved to
left-hand sidebar

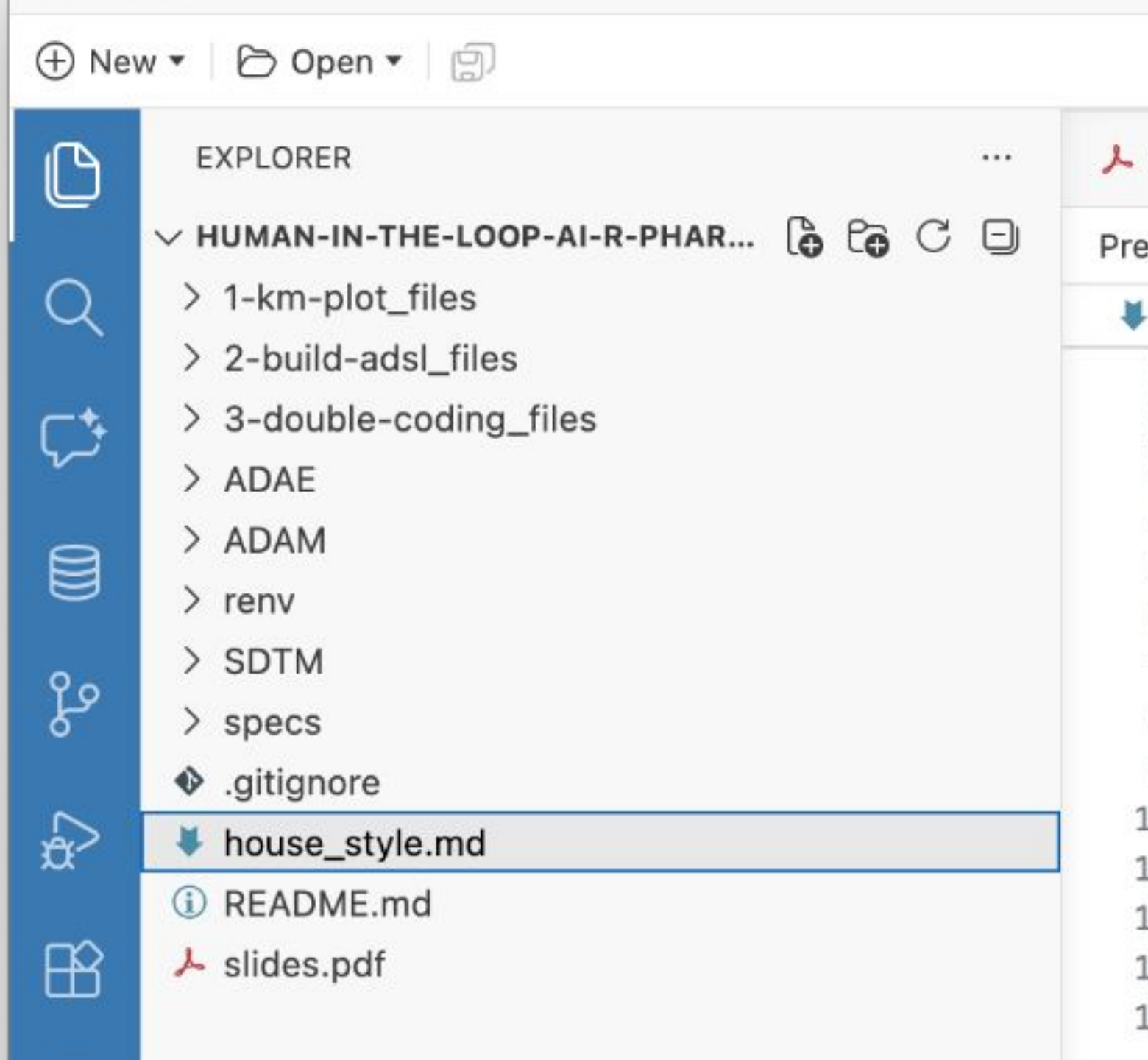


The image shows the RStudio interface. On the left is a blue sidebar with icons for various RStudio functions: a computer with a checkmark, a flask, a diamond with a plus, an upward arrow, a GitHub logo, a cube, and a three-dot menu. To the right of the sidebar is a list of installed R packages. The list includes:

- codetools** 0.2-20: Code analysis tools for R.
- compiler** 4.4.0: Byte code compiler for R.
- datasets** 4.4.0: Base R datasets.
- foreign** 0.8-86: Reading and writing data stored by some versions of 'Epi Info', 'Minita...'
- graphics** 4.4.0: R functions for base graphics.

Each package entry has a status icon (a circle with a checkmark for installed, or an empty circle for not installed) and a version number. The 'Update' button is visible next to the 'foreign' package. On the far right, a partial view of the R console is visible, showing the prompt 'R 4...' and some text.

Same with
Files



A New Data Explorer

Data: ADSL.csv — human-in-the-loop-ai-r-pharma-2026

< >

Search

🔍 🔗 🌐

Data: ADSL.csv ✕

↕ Clear Column Sorting

📂 Import Data

⚙ File Options

📄 Open as Plain Text File

☰ +

Sort by Original ▾

Filter

<

> A STUDYID

0%

> A USUBJID

0%

> # SUBJID

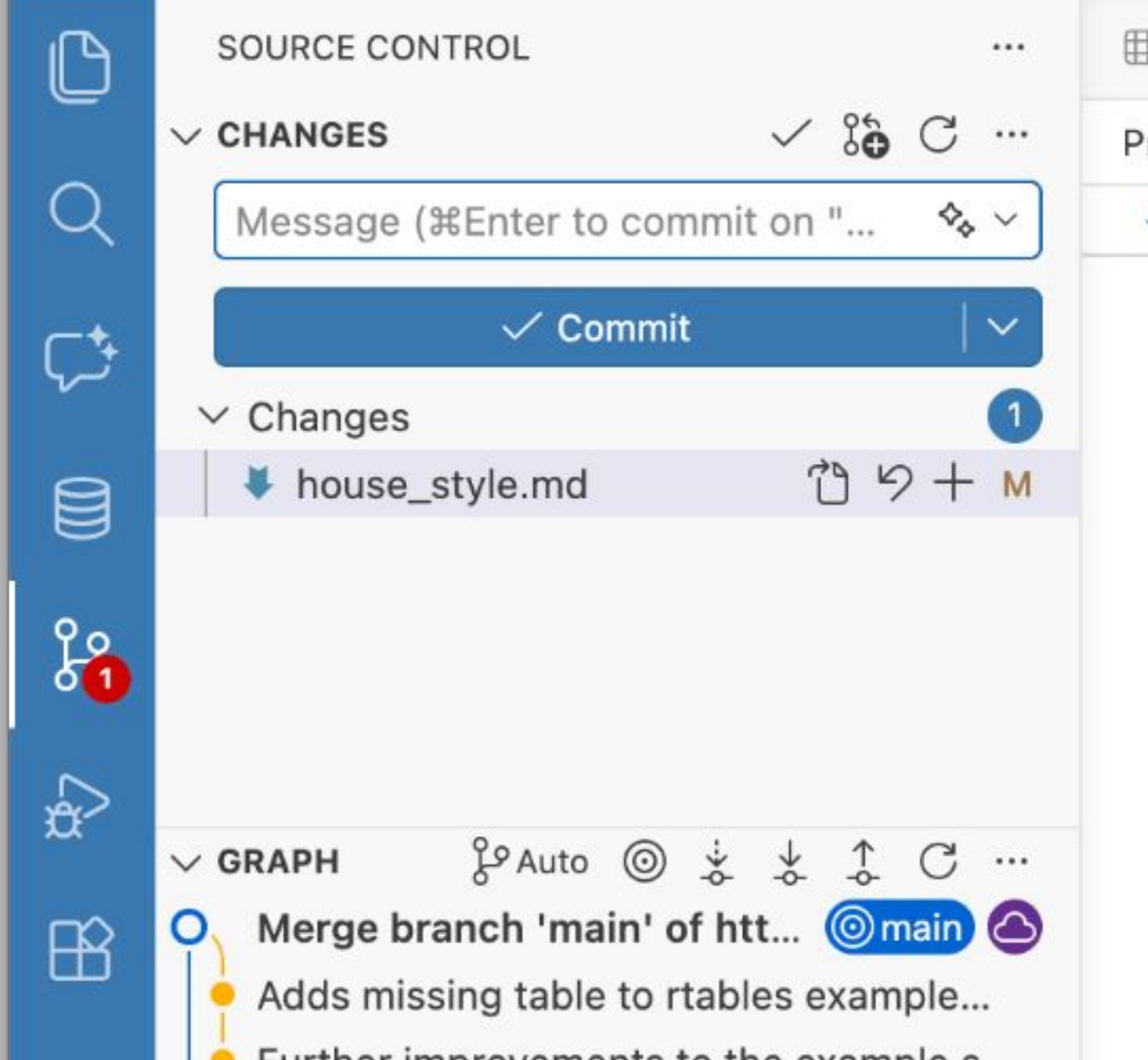
0%

> # SITEID

0%

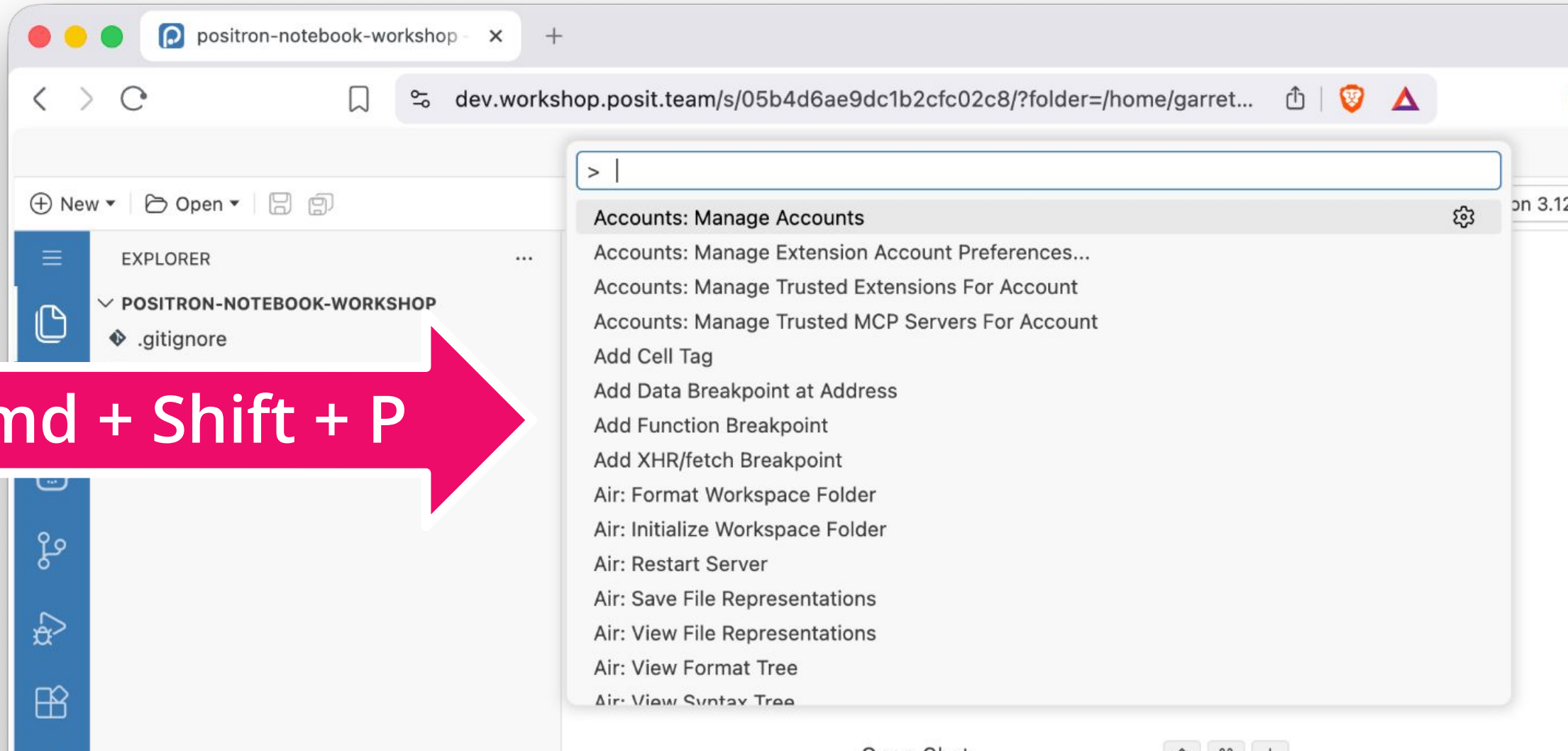
	STUDYID VARCHAR	USUBJID VARCHAR	SUBJID BIGINT
0	CDISCPIL0T01	01-701-1015	1015
1	CDISCPIL0T01	01-701-1023	1023
2	CDISCPIL0T01	01-701-1028	1028
3	CDISCPIL0T01	01-701-1033	1033
4	CDISCPIL0T01	01-701-1034	1034
5	CDISCPIL0T01	01-701-1047	1047

Best in class Git Client



Command Palette

Searchable list of all commands



Ctrl/Cmd + Shift + P

Extensions

Plug-ins for software, features, etc.



Project Manager

386K ⭐ 5

Easily switch between projects

alefragnani

Install



Git Project Manager

75K

Allows you to change easily between gi...

felipecaputo

Install



UnodeAi — Organized AI Wor...

24K

Turn AI agents into an organized workfo...

unode

Install



Giga AI Context Manager

23K ⭐ 4

AI-powered project manager that analy...

giga-ai

Install



Codex Project Manager

7K

An extension to help users initialize diff...

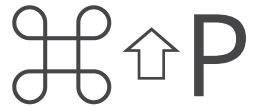
Codex Translation Editor

Install

Open the Command Palette
with **Cmd/Ctrl + Shift + P**.

Find and run the command
Developer: Reload Window.

Migrating from RStudio IDE



Preferences: Open User Settings

Settings

User Workspace

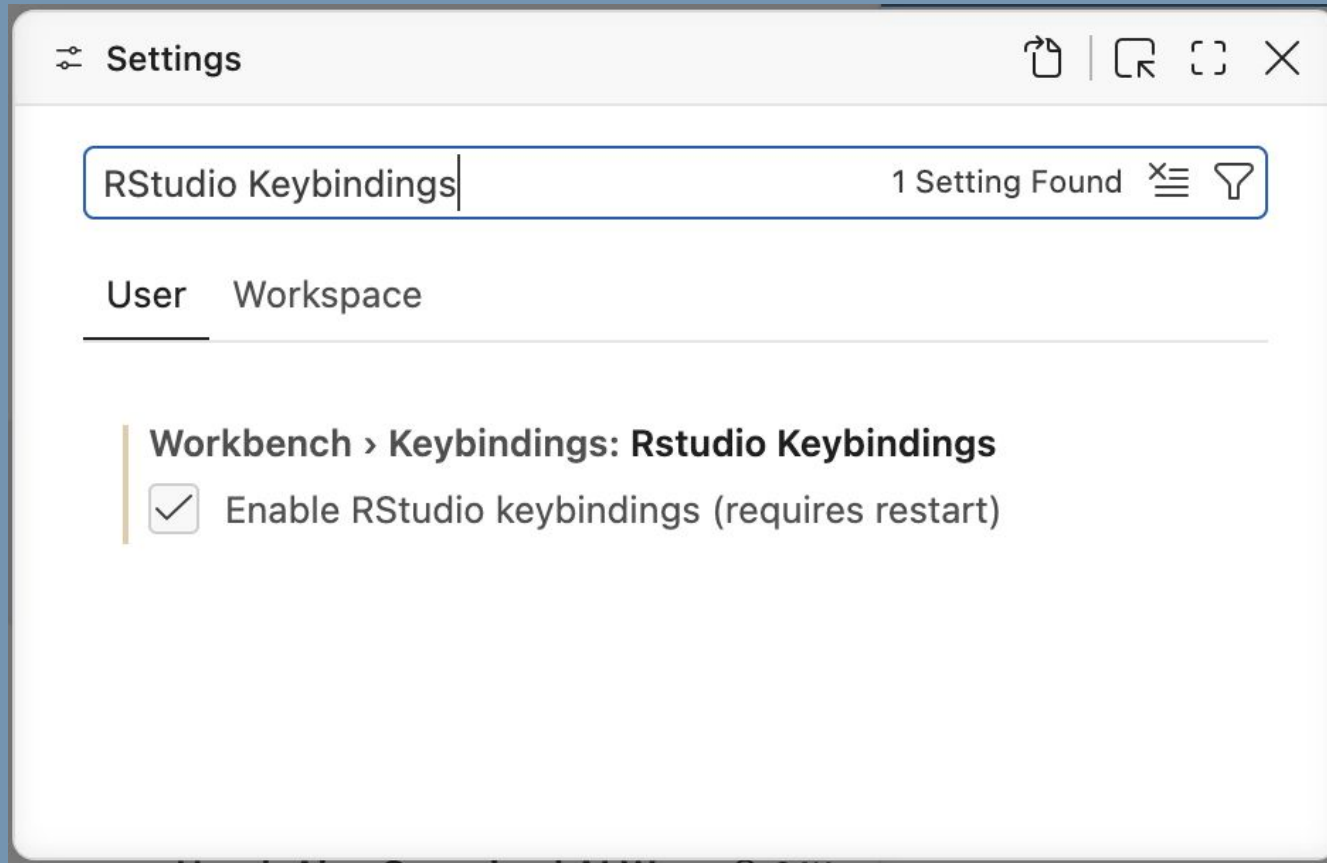
Commonly Used

- > Text Editor
- > Workbench
- > Window
- > Chat
- > Features
- > Application

Commonly Used

Editor: Font Size

Controls the font size in pixels.



Imports
RStudio's
keyboard
shortcuts

Setting

RStudio Keybindings

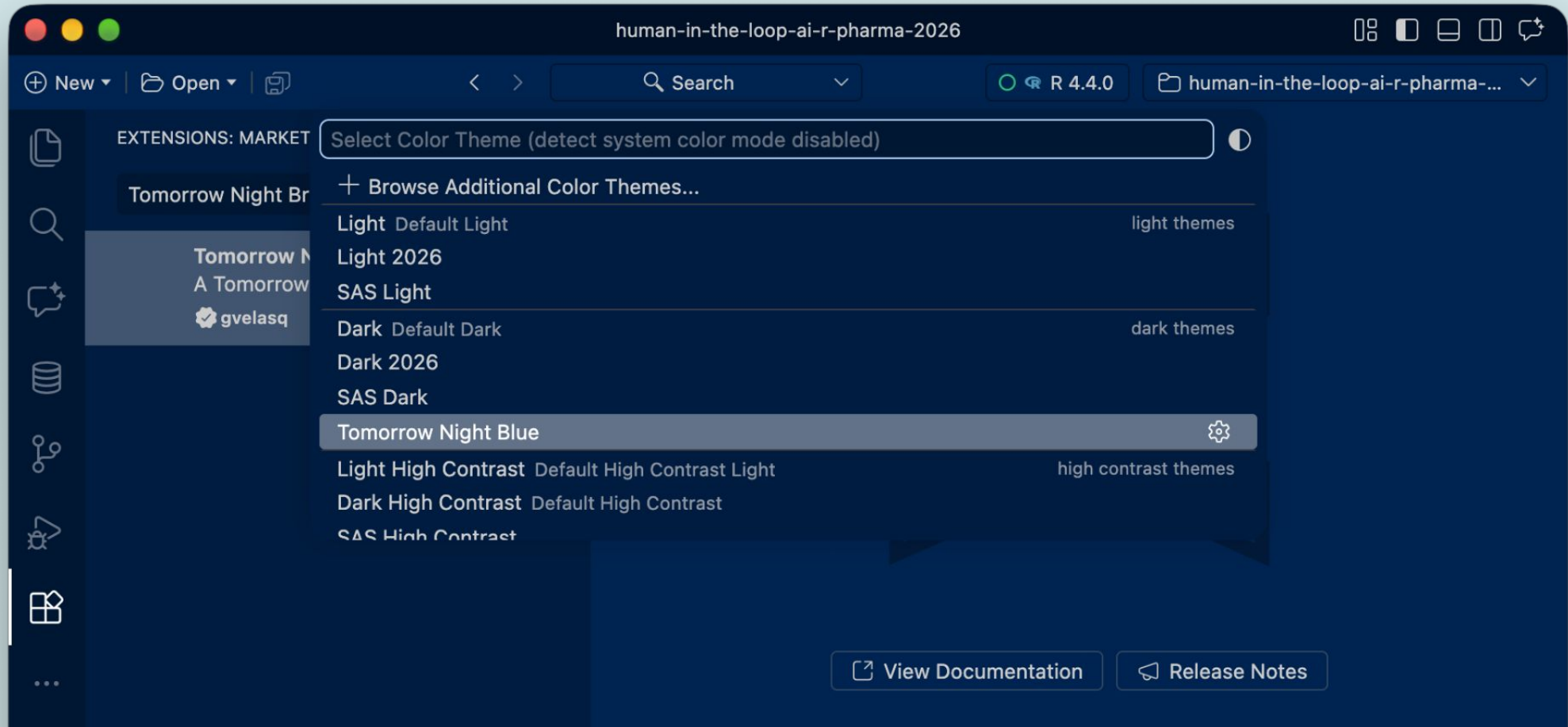
Use the Command Palette (⇧**P**) to
run **Preferences: Open User
Settings.**

Find and enable
RStudio Keybindings.

Themes

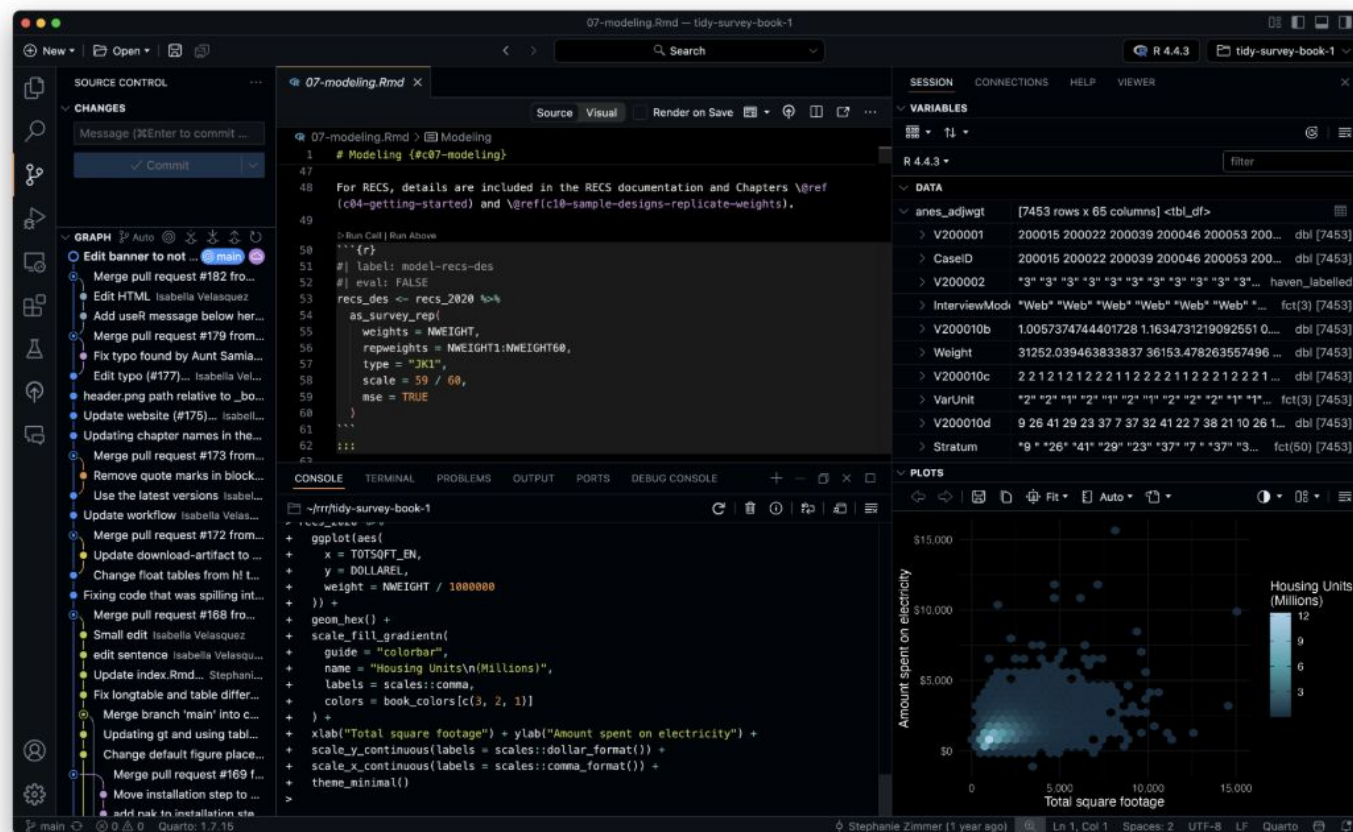


Preferences: Color Theme



Tomorrow Night Bright

An extension with RStudio's dark mode



Formatting

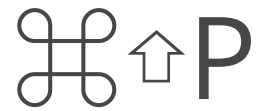
Air to Format code

Ensure that whitespace,
newlines and punctuation
conform to standards



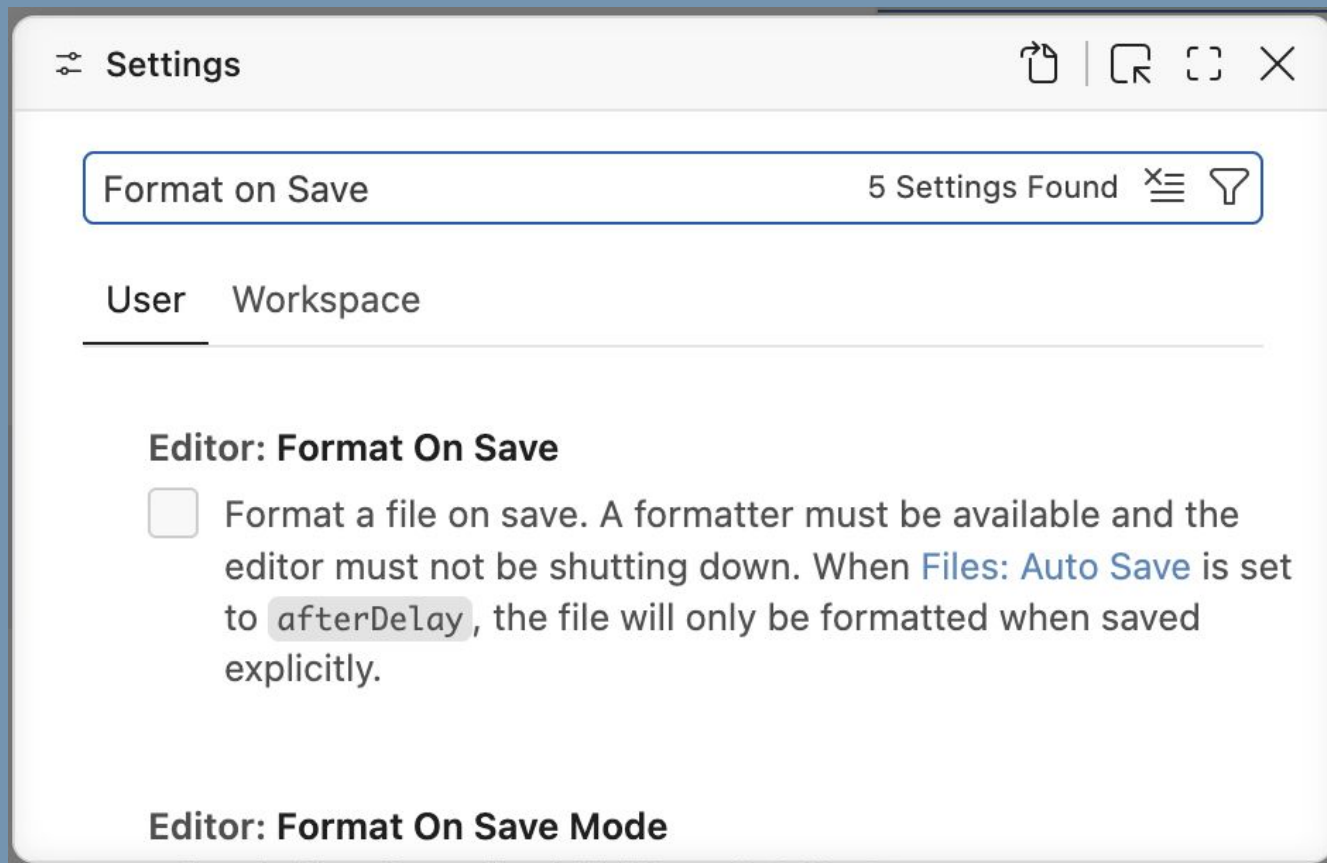


Format Document



Format Selection

```
AE_table_rtables.R 3 X  
▶ ▾ 📁 ⚠ 1 missing package ▾ 📄 📌 ...  
ADAE > AE_table_rtables.R > ...  
17 ✨  
18 suppressMessages({  
19   ·· library(rtables)  
20   ·· library(dplyr)  
21 })
```



Automatically format code

Setting	Format on Save
---------	----------------

Code Snippets

Code Snippets

follow [VS Code Syntax](#)
instead of RStudio's

```
if (condition) {  
  }  
}
```

Open an R Script with
New > New File > R File.

Type **for**. Press tab to find and use
the code snippet.

for

 **for**

[keyword]

 **for**

Insert `for` loop

{ } **for**cats::

{ } **for**foreign::

Addins



R: Run RStudio Addin

Select an RStudio Addin

Value to clipboard

clipr

Copies the results of a selected expression to the system clipboard

Output to clipboard

Copies the console output of a selected expression to the system clipboard

Run a test file

devtools

Run the current test file, using ``devtools::test_active_file()``.

Report test coverage for a file

Calculate and report test coverage for the current test file, using ``devtools::test_coverage_...``

Report test coverage for a package

Calculate and report the test coverage for the current package, using ``devtools::test_coverage_...``

Debugger mode

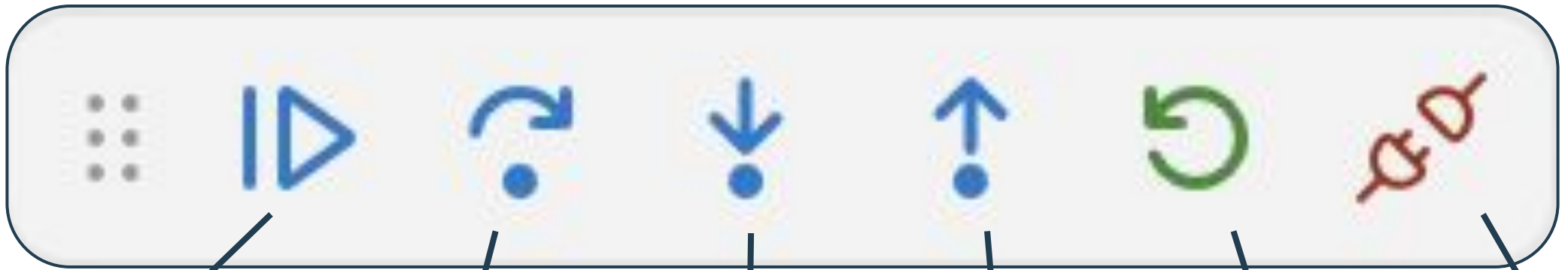
Click to add breakpoints

Positron will open
pause execution and
open a browser at
the breakpoint



```
43
44 # Build the rtables object
45 mk_row <- function(term) {
46   vals <- counts %>% filter(
47     Tmt_cell(vals$n[1],
48     fmt_cell(vals$n[2],
49     fmt_cell(vals$n[3],
50   )
51 }
52
53 hdr <- rheader(
54   rrow("", paste0(arms$TRT
55   paste0(arms$TRT
56   paste0(arms$TRT
57 )
58
59 result <- rtable(
60   header = hdr,
61   rows = lapply(top_terms
```

Click to add a breakpoint



Resume

Step
Forward

Step
Into

Step
Out

Restart

Exit

AI with Posit Assistant



Use your preferred LLM

Configure LLM Providers



MODEL PROVIDERS



Amazon Bedrock

Access Claude and other models via AWS

+ Connect



Microsoft Foundry

Access Azure OpenAI and AI Studio models

+ Connect



GitHub Copilot Preview

AI models via GitHub Copilot subscription

+ Connect



Posit Assistant

An

AI client

within Positron.



1. Adds skills
2. Shares R sessions
3. WEAR loop

Documentation at

assistant.posit.co

The screenshot displays the Posit Assistant web application in a browser window. The browser's address bar shows the URL `assistant.posit.co`. The application header includes the Posit Assistant logo, navigation links for `DOCS`, `GUIDES`, and `DOWNLOADS`, and a search bar containing the text `%K`.

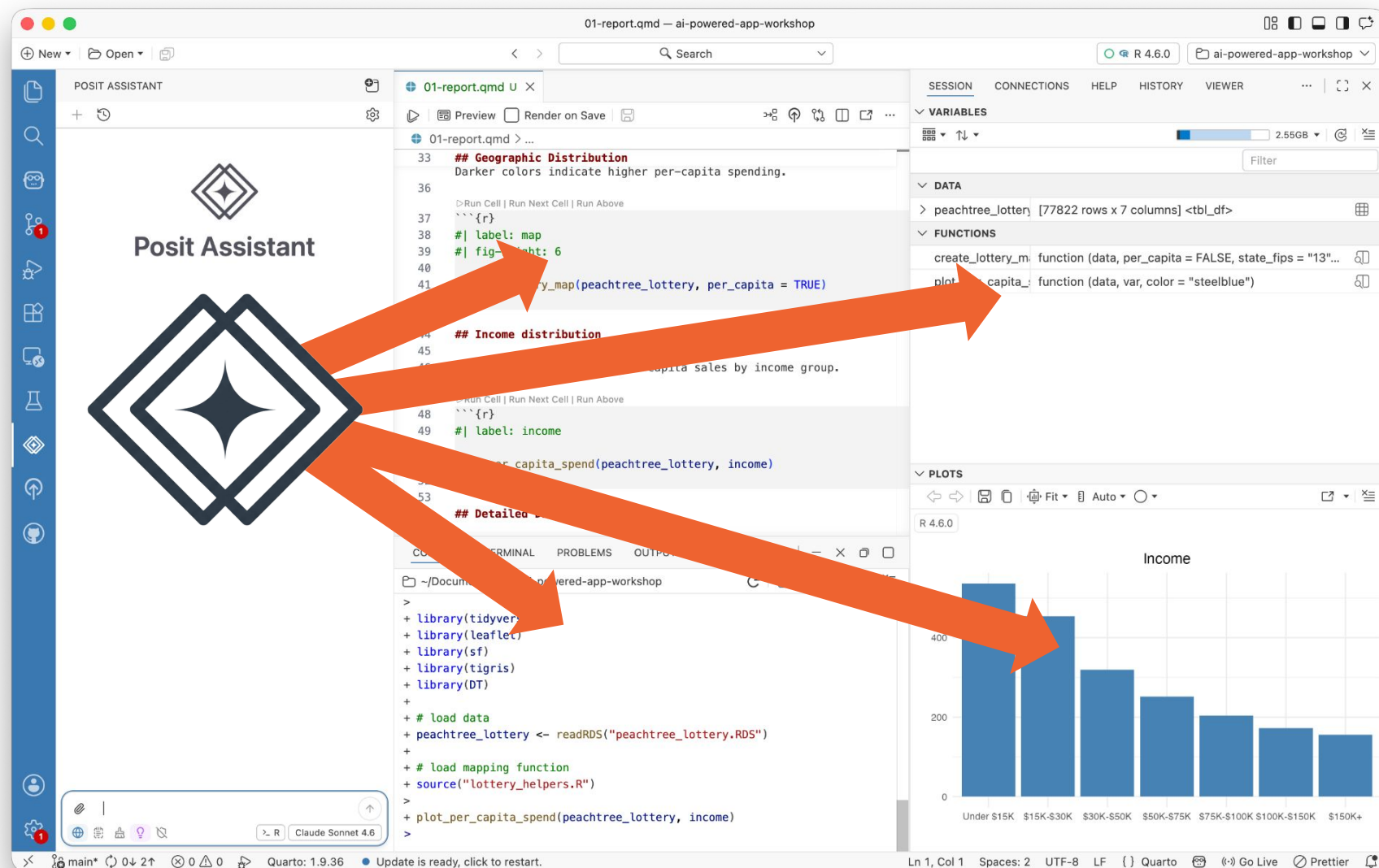
The main content area features a large heading `DOCUMENTATION` followed by the title `Posit Assistant` and the tagline `The agentic harness for data science.` Below this are two prominent buttons: `GET STARTED` and `CONFIGURATION`.

The lower portion of the page is a detailed documentation page for the `penguins` dataset. It includes a scatter plot of `bill length (mm)` versus `bill depth (mm)`. The text describes the dataset structure: 344 penguins, 3 species (Adelie, Chinstrap, Gentoo), across 3 islands, with 8 variables and 11 missing values. It provides species-specific notes, such as Chinstrap being found only on Dream Island and Adelie spreading across all three. A section on 'Between-species differences' lists characteristics like Gentoo being the largest birds and Chinstrap having the longest bills. A scatter plot is also shown with the caption: 'The scatter plot shows something interesting: there's an apparent negative correlation between bill length and bill depth when you look at the full dataset, but within each species the relationship is actually flat or slightly positive. This is a classic example of Simpson's paradox — species composition drives the overall pattern.'

On the right side of the documentation page, there is a console window showing R code and its output. The code includes `library(tidyverse)`, `data <- read_csv('penguins.csv')`, and `glimpse(data)`. The output shows the first few rows of the dataset, including columns for `species`, `island`, `bill_length_mm`, `bill_depth_mm`, `flipper_length_mm`, `body_mass_g`, `sex`, and `year`. Below the console, there is a 'Bill dimensions by species' plot showing `bill depth (mm)` versus `bill length (mm)` for the three species.

Open Posit Assistant.

Confirm that you are connected
to a model provider.



Quiz

Does this code return an error?

```
hist(cases$amount)
```

This enables
a loop:
Run code.
See results.

The screenshot displays the Posit Assistant interface within a web browser. The main window is titled "01-report.qmd — ai-powered-app-workshop". The interface is divided into several panels:

- Left Panel:** Contains the "Posit Assistant" logo and a vertical toolbar with various icons.
- Center Panel:** Displays R code in a file named "01-report.qmd". The code includes comments and function calls related to data analysis and plotting. A large, stylized diamond icon is overlaid on the code, with four orange arrows pointing from it to different parts of the code: one to the "Geographic Distribution" section, one to the "Income distribution" section, one to the "Detailed" section, and one to the "load data" section. A pink arrow points from the diamond icon to the "load data" section.
- Right Panel:** Contains a sidebar with tabs for "SESSION", "CONNECTIONS", "HELP", "HISTORY", and "VIEWER". The "SESSION" tab is active, showing a list of variables and functions. Below this, there is a "PLOTS" section displaying a bar chart titled "Income".
- Bottom Panel:** Contains a terminal window showing the output of the R code, including library loading and data reading.

The bar chart in the "PLOTS" section shows the distribution of income groups. The x-axis represents income ranges, and the y-axis represents the count of observations.

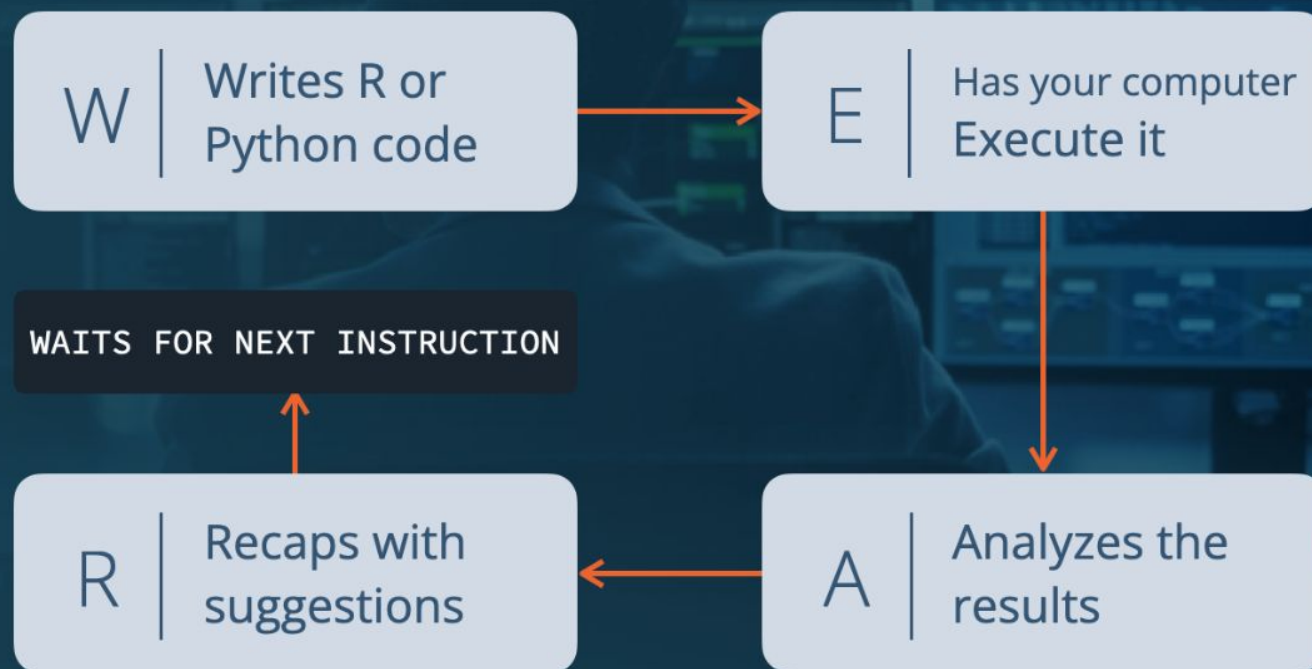
Income Group	Count
Under \$15K	400
\$15K-\$30K	350
\$30K-\$50K	300
\$50K-\$75K	250
\$75K-\$100K	200
\$100K-\$150K	150
\$150K+	100

WEAR loop

Code is the

most auditable

way to explore
data with AI



Use Posit Assistant to explore the datasets in the ADaM folder.

 **executeCode:** load both ADAE extracts and compare structure

Can Posit Assistant use executeCode?

Posit Assistant would like to run this R code to load both ADAE extracts and compare structure.

```
library(tidyverse)

prod <- read_csv("ADAE/ADAE_prod.csv", show_col_types = FALSE)
qc <- read_csv("ADAE/ADAE_qc.csv", show_col_types = FALSE)

list(
  dim_prod = dim(prod),
  dim_qc = dim(qc)
)
```

✓ Allow

✗ Decline



Allow once

✓ Allow executeCode tools for this session
for all inputs

📁 Always allow executeCode tools for this
project
for inputs matching *

What should Posit Assistant do instead?

Ex: Please use a different plotting library.

Cancel

Decline

Allow, decline,
and edit your
LLM code
requests

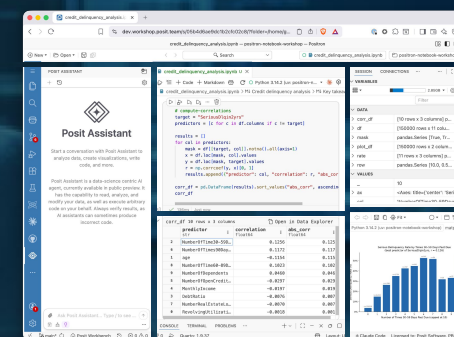
You



AI



Posit Assistant



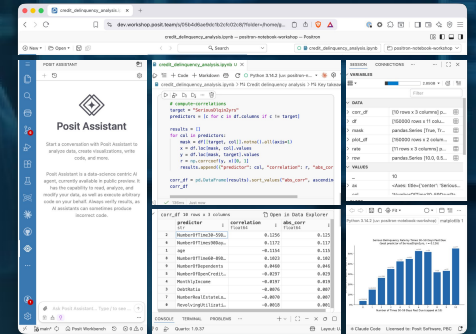
Without Positron Assistant

You are the middleman and files become the intermediary

AI

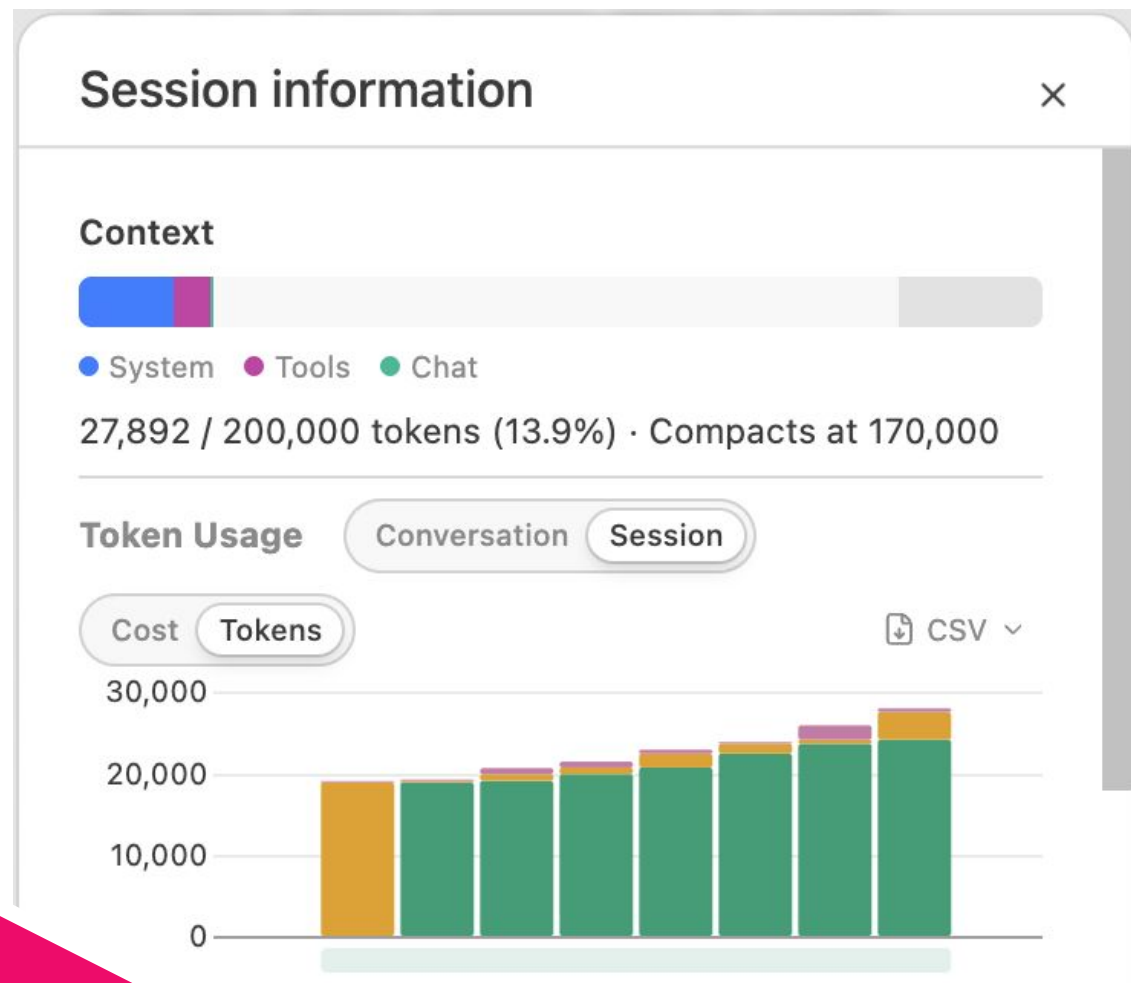
You

Files



Context, Cost, Cache, and Tokens

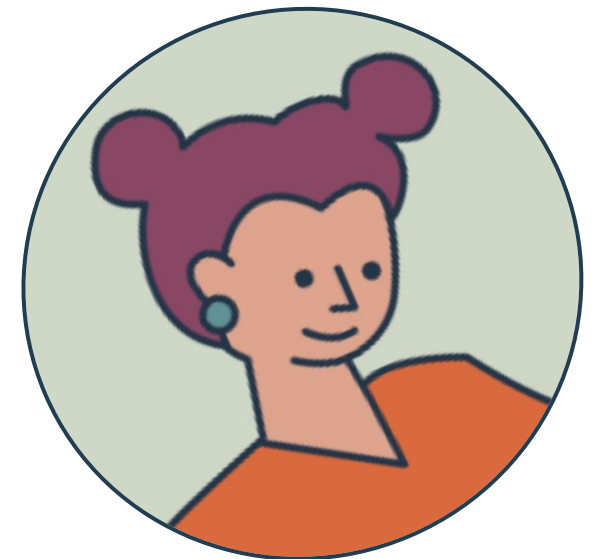
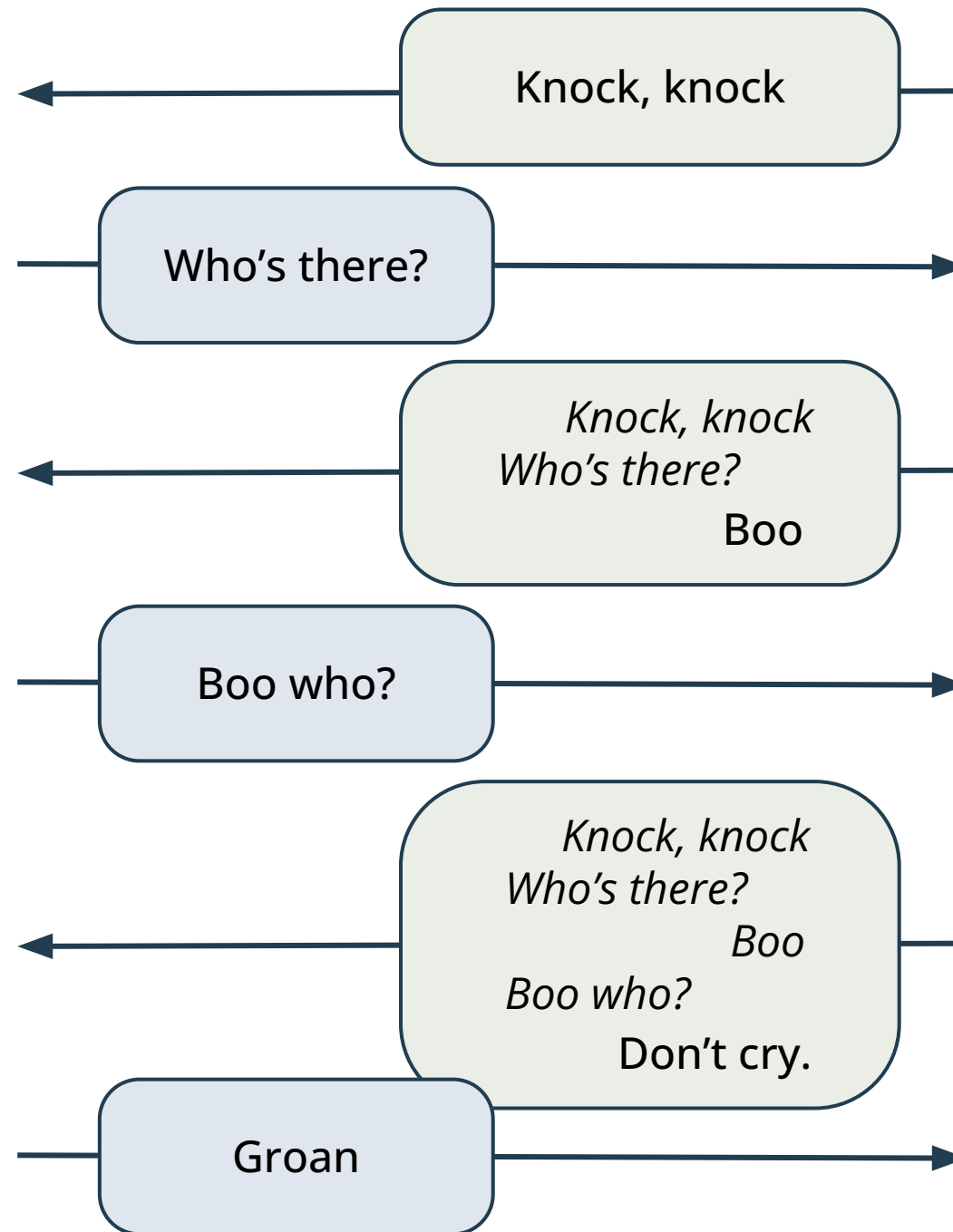
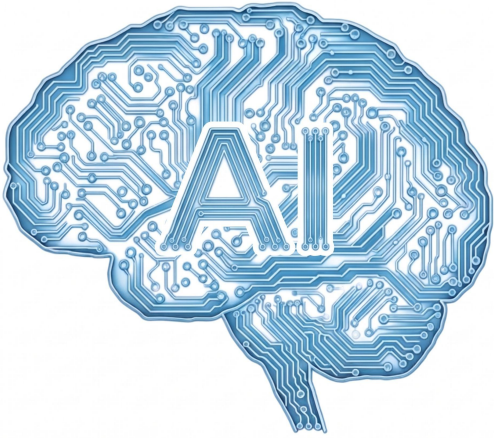
Live and transparent
monitoring



Ask Posit AI

🌐 🚫 🤖

👁️ R 🔥 30:03 ● 27K · \$0.08 ⚡ Haiku 4.5

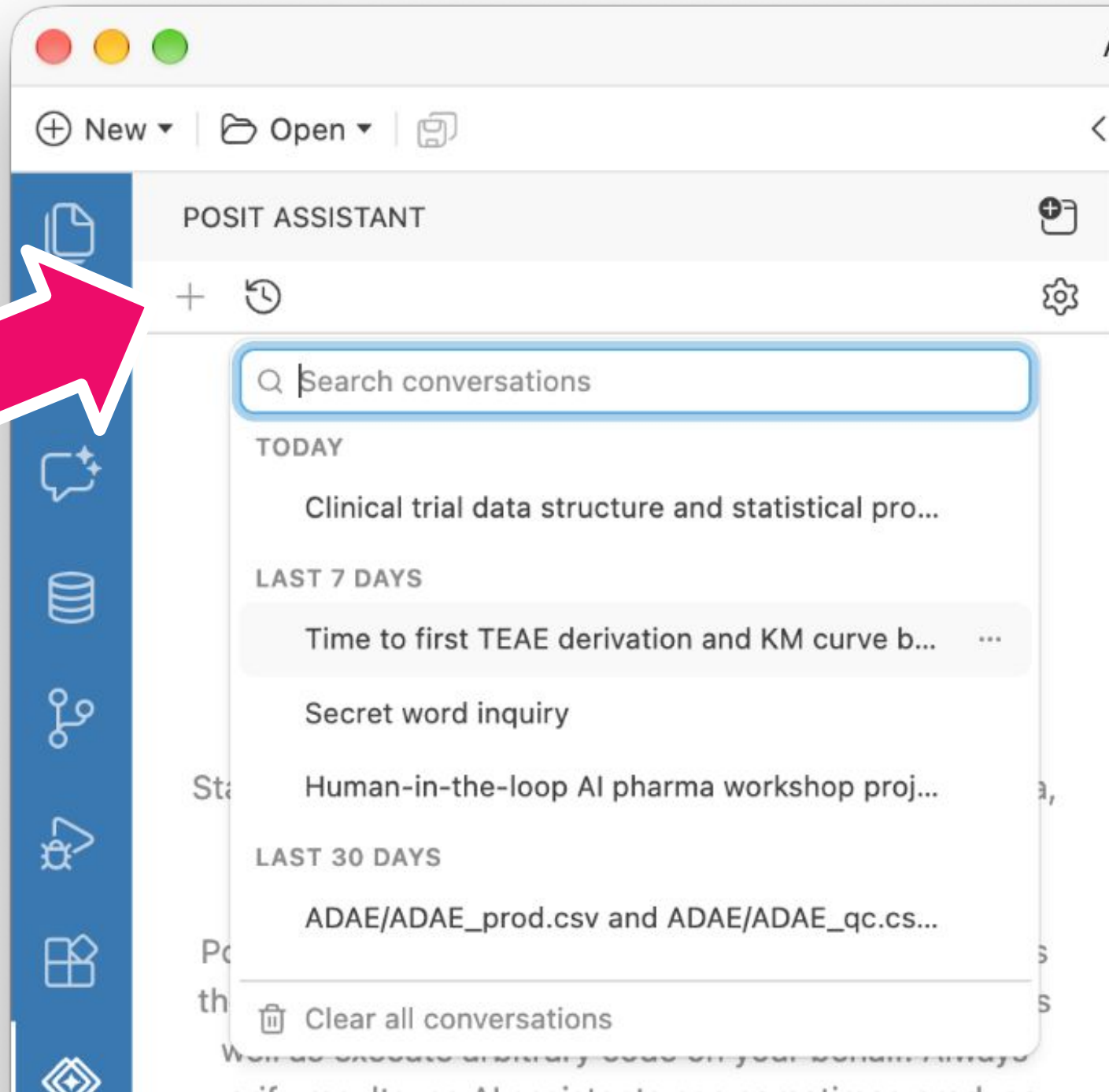


To minimize token usage

1. Select the smallest **model** for the job
2. Keep the **cache** warm if you plan to do more soon
3. Monitor **conversation length**
4. Write clear, detailed (i.e. long) **prompts**

Start a new Conversation

Or return to a previous one



Ask Posit Assistant:

Use ADAM/ADAE.csv and ADAM/ADSL.csv to derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM.

Teaching the AI



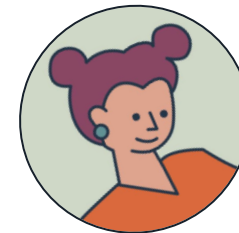
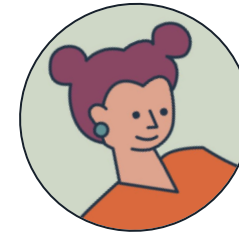
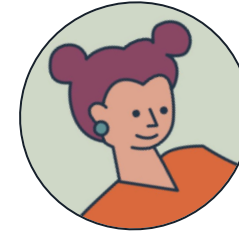
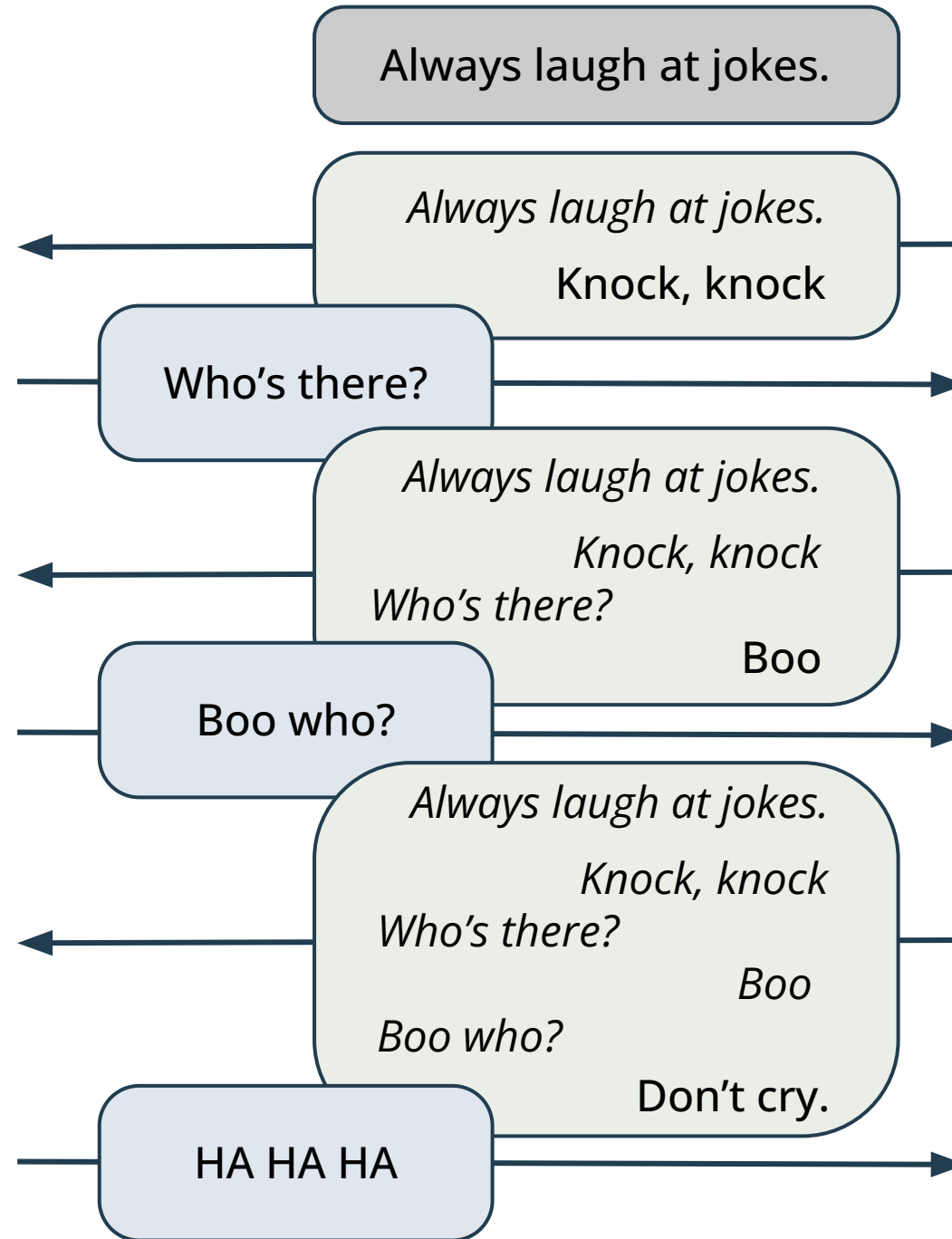
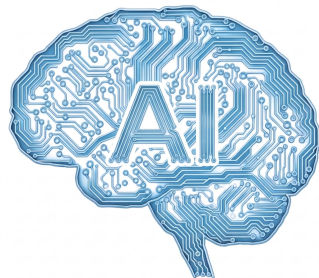
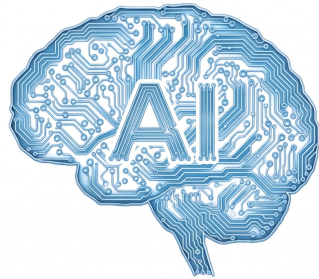
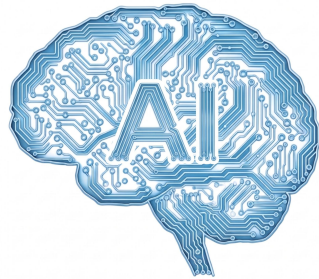
Memory (AGENTS.md)

Instructions to know for every conversation

Skills

Just-in-time instructions on specific topics

AGENTS.md



Make a file named AGENTS.md. Save this in it.

1. Kaplan-Meier figures

- **Palette** (by arm, in this fixed order): Placebo = `#BDBDBD` (grey), low dose = `#4A90D9` (blue), high dose = `#D9534F` (red). Always map arms to colors in protocol order, not alphabetical order.
- **Always include:** a number-at-risk table below the x-axis, and a 95% confidence band (shaded, 20% opacity). The risk table should be no smaller than 85% of the plot to ensure legibility.
- **Legend placement:** the legend must never overlap the plot area, since this obscures the data. Always orient the legend horizontally and place it beneath the plot.
- **Axis labels:** x-axis in days unless the audience is a general clinical one, in which case convert to months (divide by 30.4375) and say so in a footnote.
- **Title format:** `**<Endpoint> by Treatment Arm**` -- e.g. **"Progression-Free Survival by Treatment Arm"**. No dataset name, no study ID, in the title itself (those go in a caption/footnote).
- **Censoring marks:** show tick marks for censored observations; do not hide them for a "cleaner" look.
- **Y axis limits:** the y axis should always begin at 0 and end at 1.

In a new conversation, remake your plot.

Use ADAM/ADAE.csv and ADAM/ADSL.csv to derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM.

Skills

What is a skill?

Files in an agreed upon structure and location

```
my-skill/  
├── SKILL.md           # Required: skill definition with metadata  
├── scripts/          # Optional: executable scripts (R, Python, Bash)  
├── references/       # Optional: additional documentation  
└── assets/           # Optional: templates, schemas, data files
```

AGENTS.md

When you make **plots**, I want you to follow these instructions.
(Several lines of instructions that get loaded into every conversation).

When you make **tables**, I want you to follow these instructions.
(Several lines of instructions that get loaded into every...

Hi

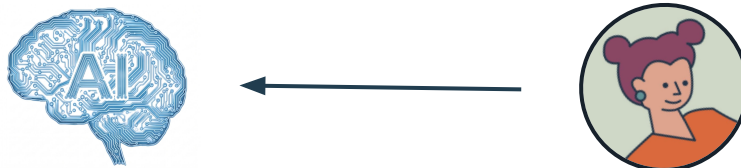


MCP

There are three tools you can use:

- **The Plot tool**
What the plot tool does. How to use the plot tool. What the plot tool returns. This takes many lines that get loaded into every conversation.
- **The Table tool**
What the tool...

Hi



Skills

These are available to read if you need them:

- How to make **plots**
- How to make **tables**

Hi



Posit makes and shares skills



A collection of skills from Posit

<https://github.com/posit-dev/skills>

In a new conversation, make a table.

Use ADAM/ADAE.csv and ADAM/ADSL.csv and the DT package to create a table of the top 10 treatment-emergent adverse events (TRTEMFL == "Y", SAFFL == "Y"). Show counts and percentages by treatment arm.

Give Posit Assistant specific instructions on how to improve the table. Consider how to format values, sort rows, label columns, etc. Make the revised table.



Create a skill with

/create-skill

Slash Commands

- `/compact` Compact conversation to reduce token usage
- `/context` Show session context and token usage
- `/clear` Start a new conversation, keeping the current context
- `/clean` Enter Cleaning mode
- `/create-skill` Create new Agent Skills following the template
- `/marketplace` Open the plugin & marketplace menu
- `/mcp` Show MCP servers and sign in to them
- `/microcompact` Micro-compact tool results in the chat

/



Claude 4 Sonnet ▾

Run

/create-skill Create a skill that makes adverse event tables like the one we just made. An LLM should be able to use the skill to provide a table in my preferred style whenever I ask for a similar table.

Collecting results





Summarize your
conversation with
the **/notebook**
and **/report**
commands



Commands

/report Create a report as a Quarto document

/notebook Create a report as a Jupyter notebook

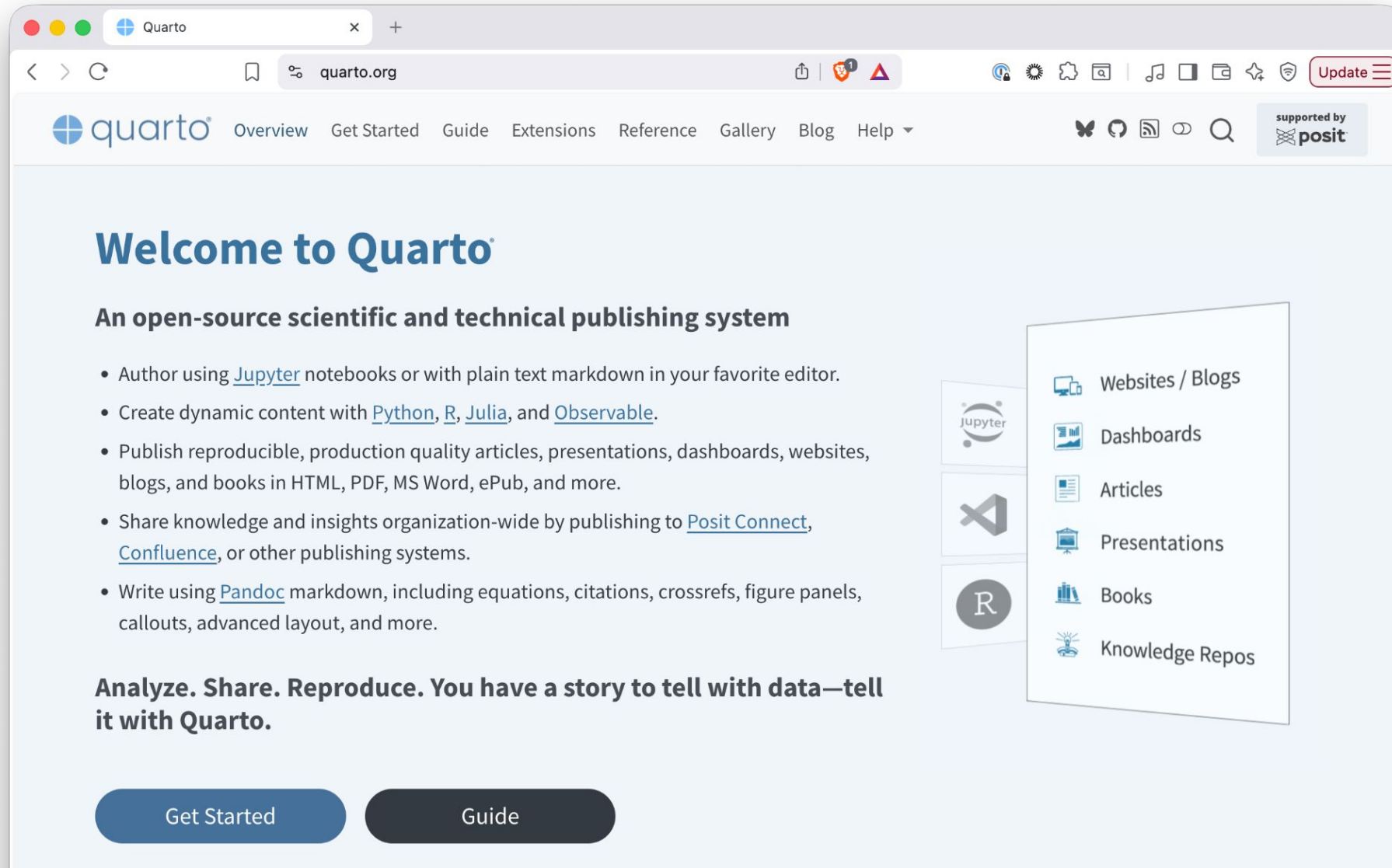
/



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quarto.org



Choose a conversation and turn it
into a .qmd with **/report**.

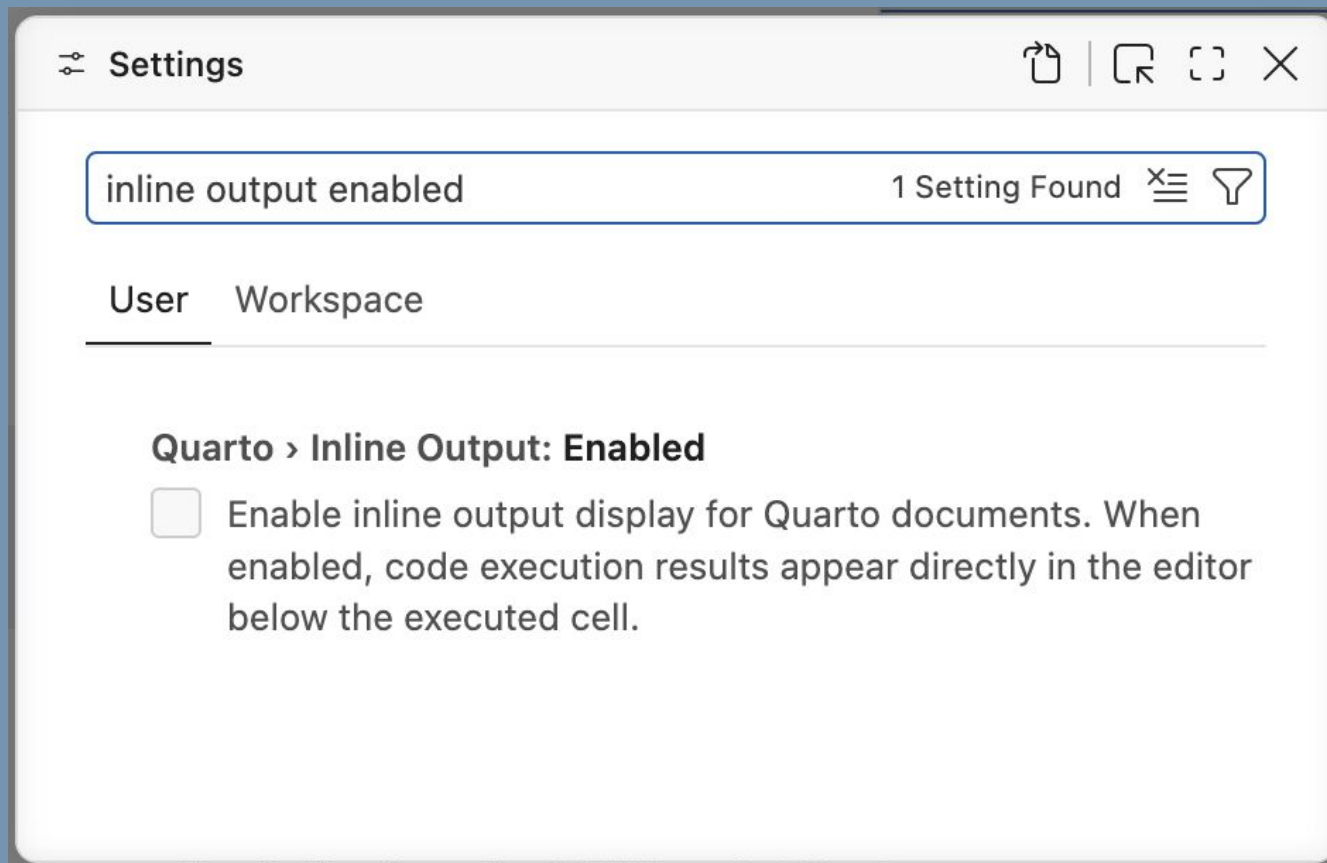
Edit the file and then click Preview.

Hides code

title: "My report"

echo: false





Turn on inline output for Quarto

Setting

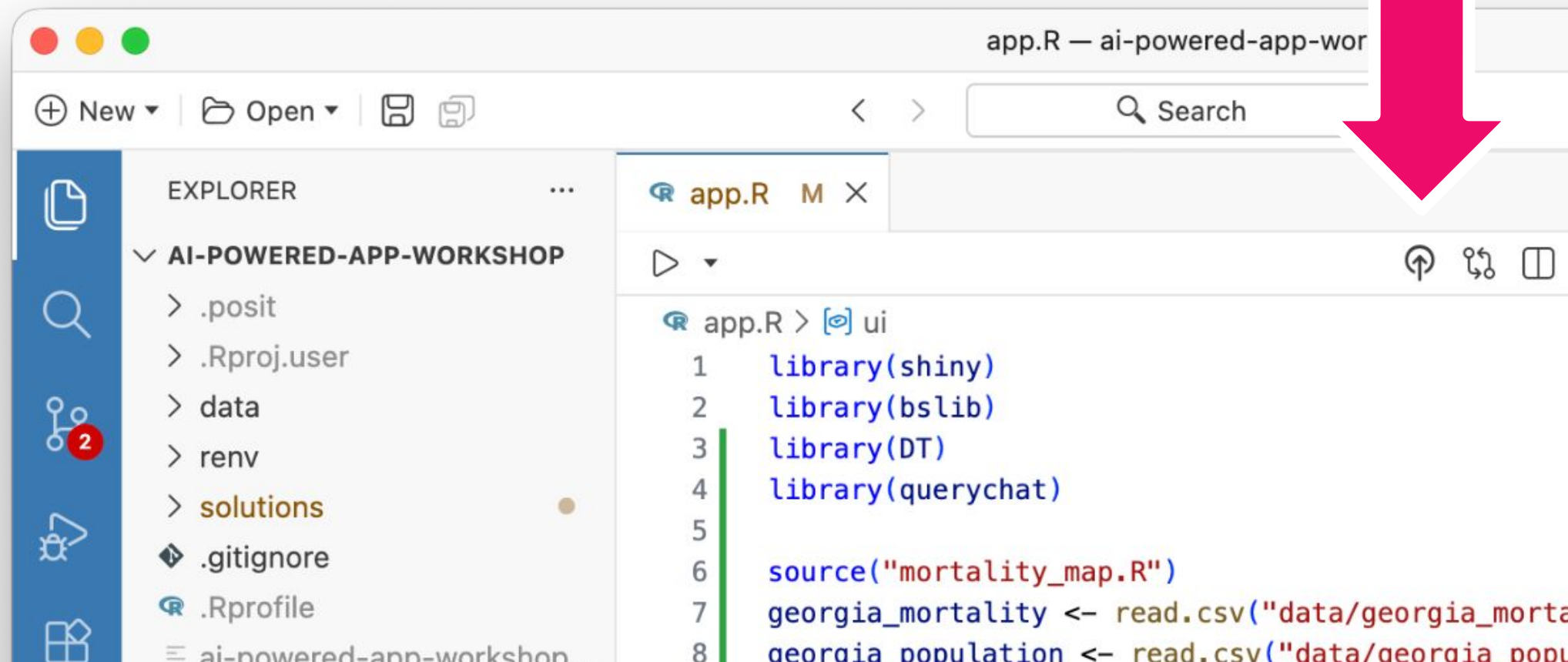
Inline Output Enabled

Publishing from Positron



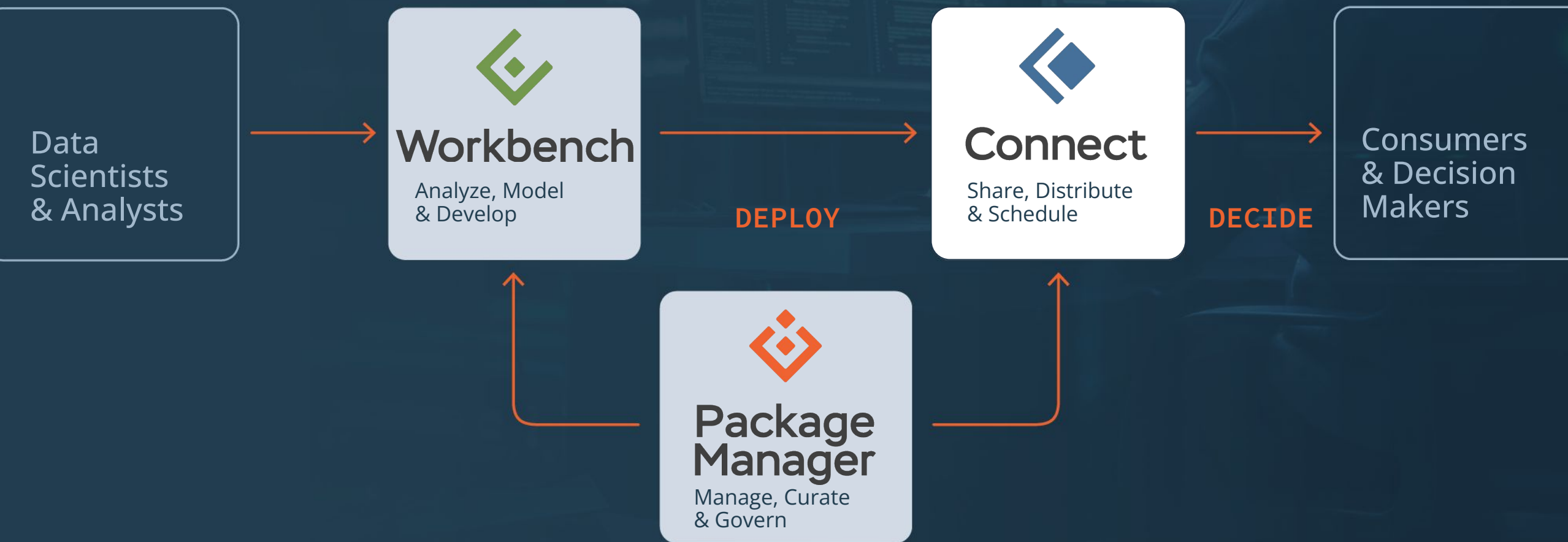


Push button publishing to Posit Connect



Posit Connect is a part of Posit Team

Posit Connect makes it easy to deploy and share data science work across teams. Publish R and Python models, dashboards, APIs, and reports to the people who need insights most—securely, seamlessly, and at enterprise scale.



Posit Academy

FREE

Self-paced, on-demand learning for every Posit user. No cost. Live now at academy.posit.co



ON-DEMAND

Product Courses

Self-paced courses covering Connect, Workbench, and Package Manager workflows. Available 24/7



HANDS-ON

Labs

Sandboxed Instruqt environments to practice real Posit workflows. No setup, no risk to production data: perfect for new users.



LIVE

Workshops

Live online sessions led by the CAPE team. Small-group, interactive format for customers who want guided practice.

FREE

Posit Academy

- On-demand courses, labs & live workshops
- Open to all customers, prospects & open source users
- Login and Learn

PAID

Academy Apprenticeships

- Cohort-based, mentor-led 6-8 week programs
- Premium upskilling in R & Python
- The same highly rated program with a new name

Please email me if you have an
example you would like to see
taught on academy.posit.co

garrett@posit.co

Thank you
What questions do you have?

Follow along with me

Open Posit Assistant ( in sidebar). Ask it to diagnose a double programming discrepancy:

ADAE/ADAE_prod.csv and ADAE/ADAE_qc.csv are two versions of an ADAE extract. They should be identical but are not. I'd like to figure out the cause. Find every discrepancy, and tell me if there's a pattern to which records disagree.

Follow along with me

Open Posit Assistant ( in sidebar). Ask it to diagnose a double programming discrepancy:

ADAE/ADAE_prod.csv and ADAE/ADAE_qc.csv are two versions of an ADAE extract. They should be identical but are not. I'd like to figure out the cause. Find every discrepancy, and tell me if there's a pattern to which records disagree.

Your Turn

Click + (top left) to start a new conversation.

Then ask Posit Assistant to build an ADSL table:

Build an ADSL table from the SDTM files in SDTM/ using the specification in specs/ADSL_spec.xlsx. Use {metacore} to read the spec, {admiral} to derive the variables it describes, {metatools} to check the result against spec, and {xportr} to export a submission-ready .xpt.

Follow up as needed.

Your Turn

Click + (top left) to start a new conversation.

Then ask Posit Assistant to build an ADSL table:

Build an ADSL table from the SDTM files in SDTM/ using the specification in specs/ADSL_spec.xlsx. Use {metacore} to read the spec, {admiral} to derive the variables it describes, {metatools} to check the result against spec, and {xportr} to export a submission-ready .xpt.

Follow up as needed.

Your Turn

Run `/report` in Positron Assistant to save your code in an editable format.

Then click **Preview** at the top of the .qmd file

Your Turn

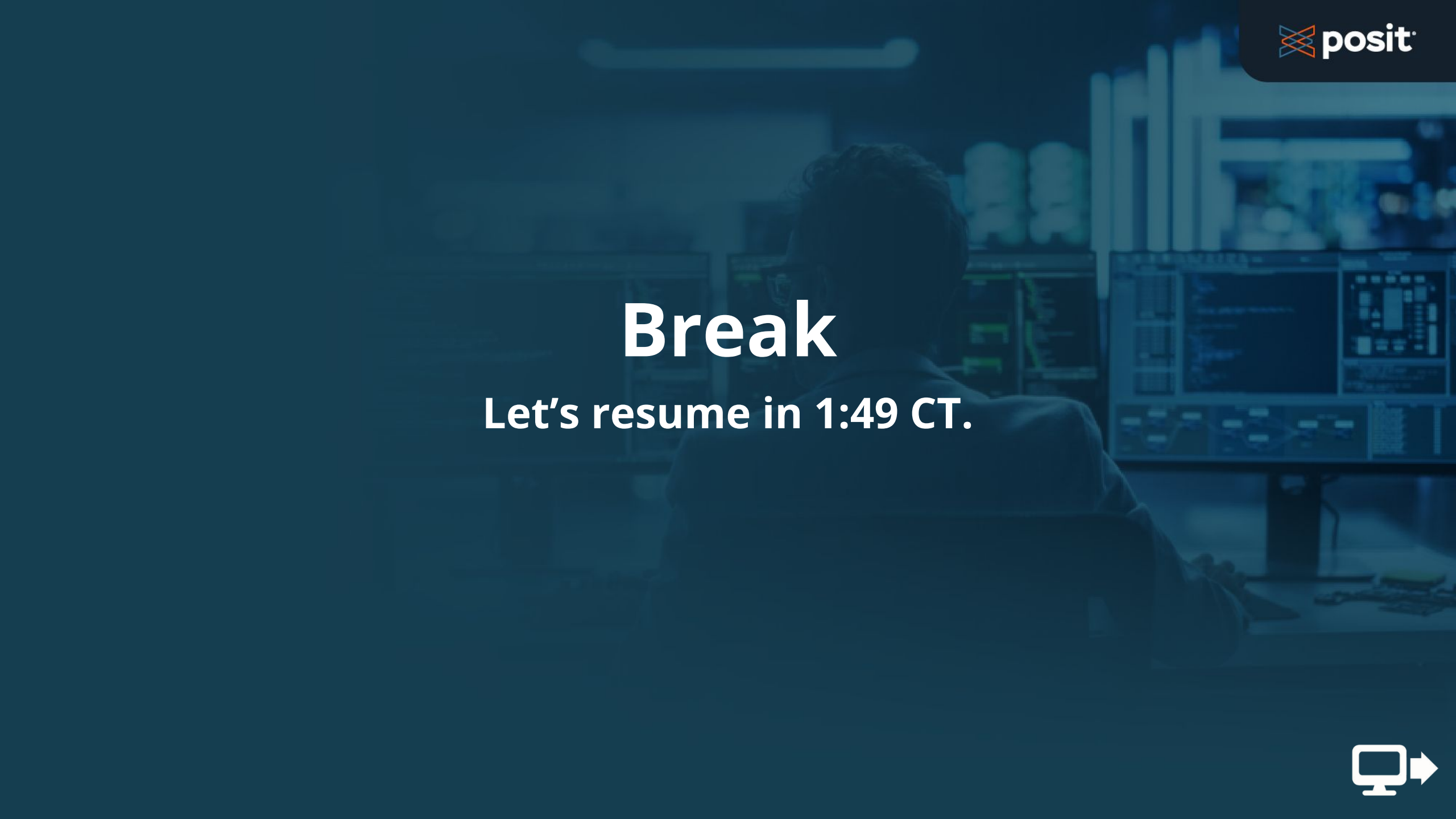
Run `/report` in Positron Assistant to save your code in an editable format.

Then click **Preview** at the top of the .qmd file



Demo - Quarto reports for R and Python.



The background of the slide is a dark, blue-tinted image of a person sitting at a desk with multiple computer monitors. The person is seen from the back, looking at the screens. The screens display various data visualizations, including charts and graphs. The overall atmosphere is professional and tech-oriented.

Break

Let's resume in 1:49 CT.



Your Turn

Start a new conversation.

ADAE/AE_table_rtables.R creates a working AE frequency table built with {rtables}. Ask Posit Assistant to write an R script that rebuilds the exact same table, but instead uses {cards} and {tfrmt}.



Your Turn

Start a new conversation.

ADAE/AE_table_rtables.R creates a working AE frequency table built with {rtables}. Ask Posit Assistant to write an R script that rebuilds the exact same table, but instead uses {cards} and {tfrmt}.

Your Turn

Start a new conversation and ask Posit Assistant to make a KM plot.

Using ADAM/ADAE.csv and ADAM/ADSL.csv, derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM. For this task, ignore the style guide in house_style.md.



Your Turn

Start a new conversation and ask Posit Assistant to make a KM plot.

Using ADAM/ADAE.csv and ADAM/ADSL.csv, derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM. For this task, ignore the style guide in house_style.md.

Your Turn

Rename house_style.md to AGENTS.md.

Start a new conversation.

Run this prompt that does not mention style.

Using ADAM/ADAE.csv and ADAM/ADSL.csv, derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM.

Your Turn

Rename house_style.md to AGENTS.md.

Start a new conversation.

Run this prompt that does not mention style.

Using ADAM/ADAE.csv and ADAM/ADSL.csv, derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM.

Your Turn

Run

/create-skill Use the information in AGENTS.md to build a skill that makes Kaplan Meier plots that match the described style. Then delete AGENTS.md.

Then, in a new conversation re-run

Using ADAM/ADAE.csv and ADAM/ADSL.csv, derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM.

